ICDFA | WADF 105 | LABORATORY 2

EVIDENCE REPORT

OPNsense Firewall Policy Testing Logging and Packet Analysis

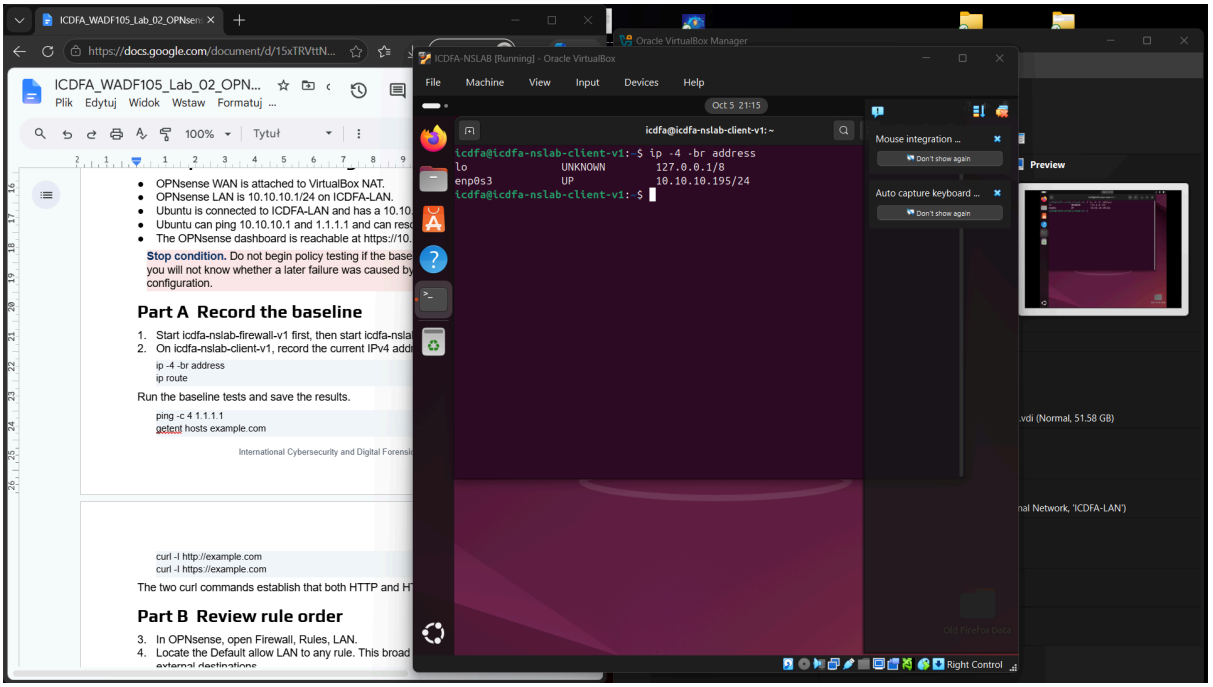
Submission Date 06 October 2026

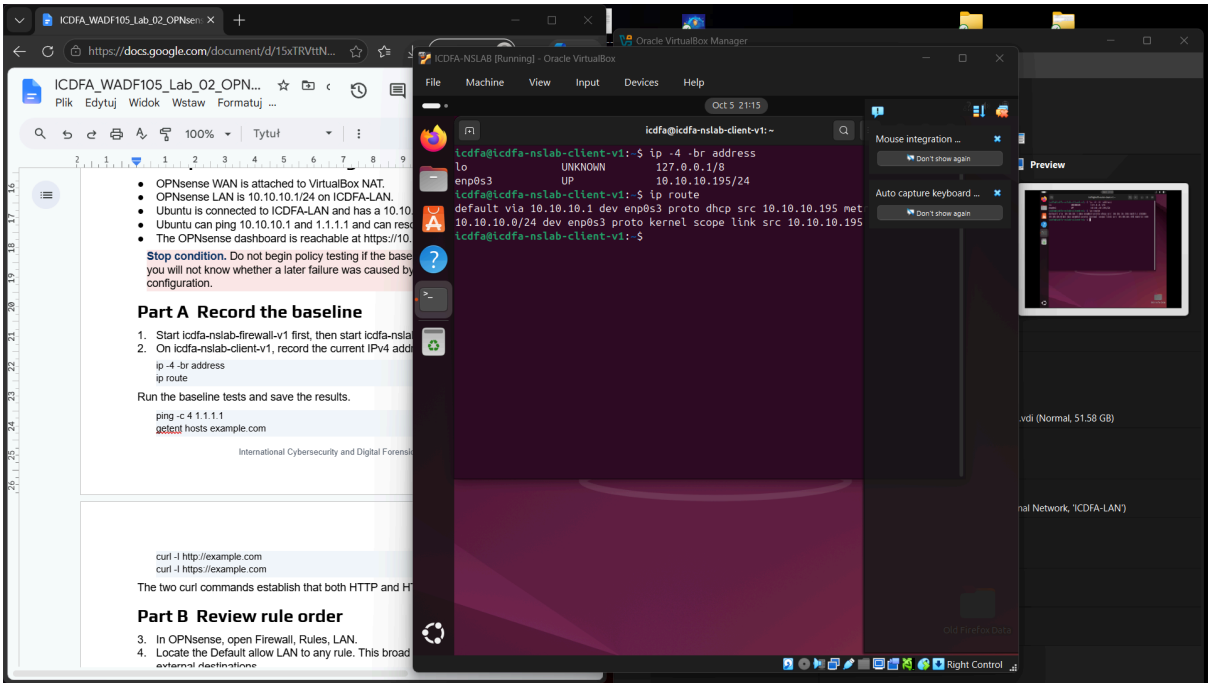
Student identification

Full name	Kenneth Chigozie Ukata
Registration number	FCDF2617034
Linux distribution / version	Kali DEBIAN LINUX
VM hostname	KaliLinuxKen

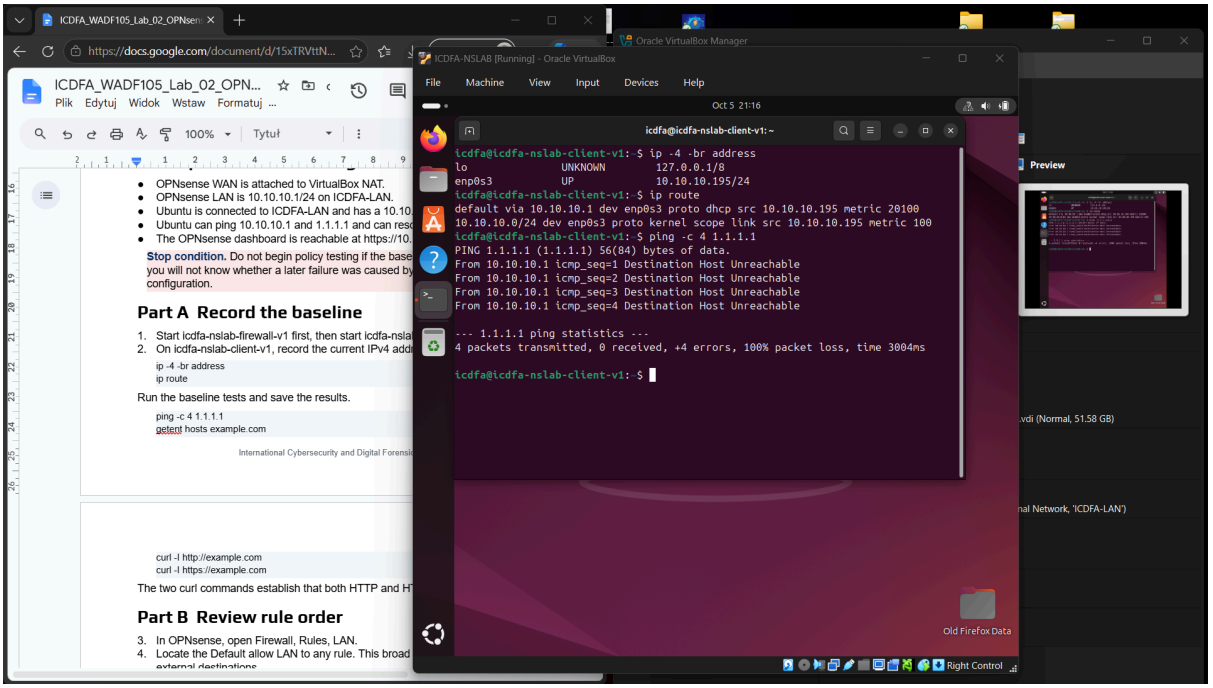
EVIDENCE:

Recording the current IPv4 address and default route
Command “ip -4 -br address”, “ip route”:





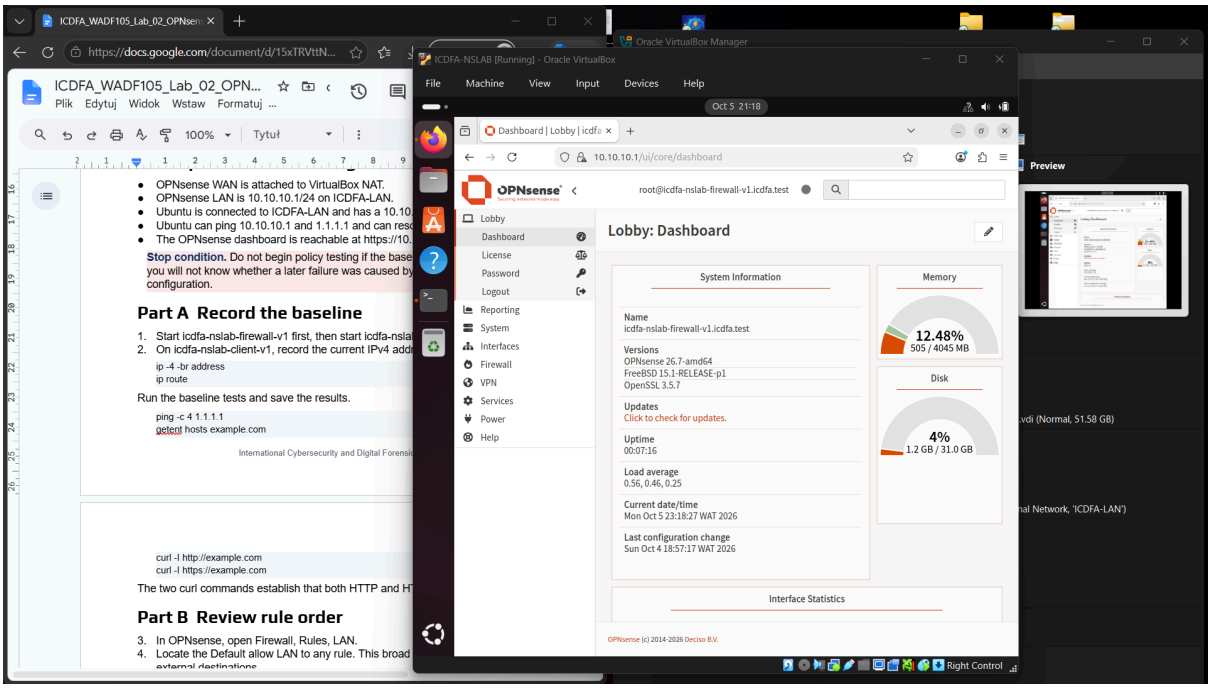
Running command “ping -c 4 1.1.1.1”

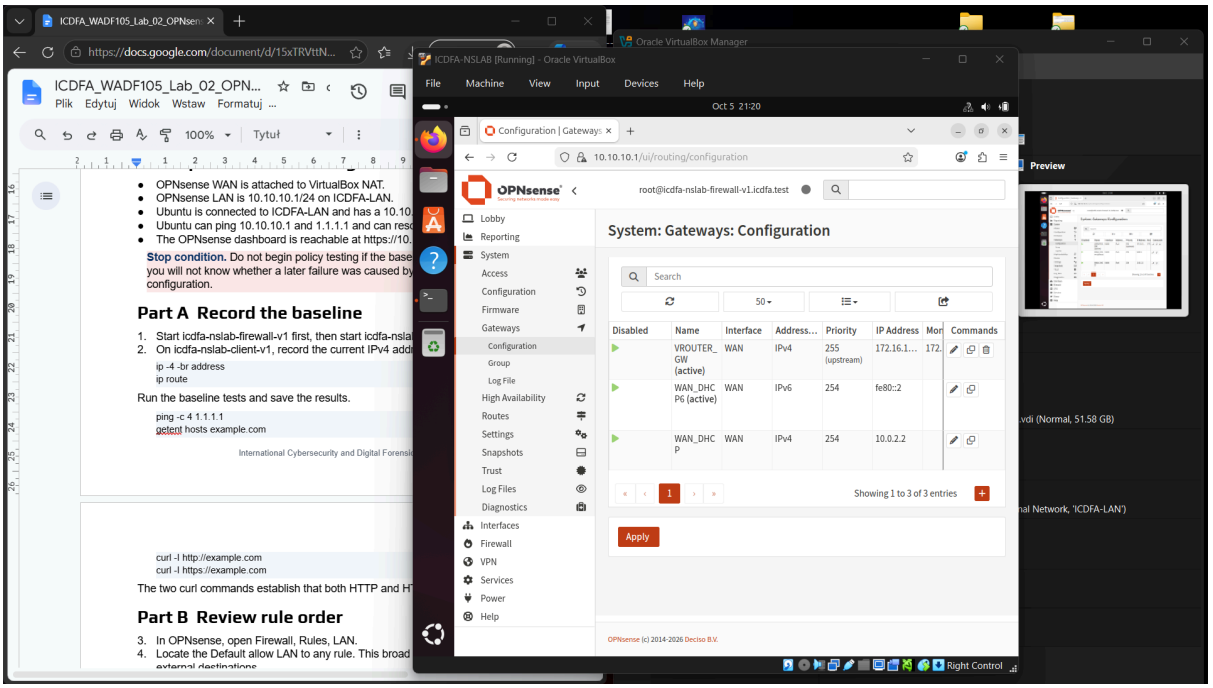


Upon starting **Lab 2**, I discovered that my previous fix for the interface issue was only temporary. Once I turned off the virtual machine and powered it back on, the system refreshed to its default state, and ping commands from my Ubuntu client failed once again with errors and 100% packet loss.

What Went Wrong: The temporary script and console changes I made in Lab 1 did not permanently save to the OPNsense system configuration. After the system rebooted, OPNsense restored its default, incorrect **LAN** and **WAN** interface assignments and wiped out my gateway settings. Because the **Upstream Gateway** was not permanently saved for

the WAN interface, the firewall dropped all outbound traffic originating from my client network (10.10.10.0/24).

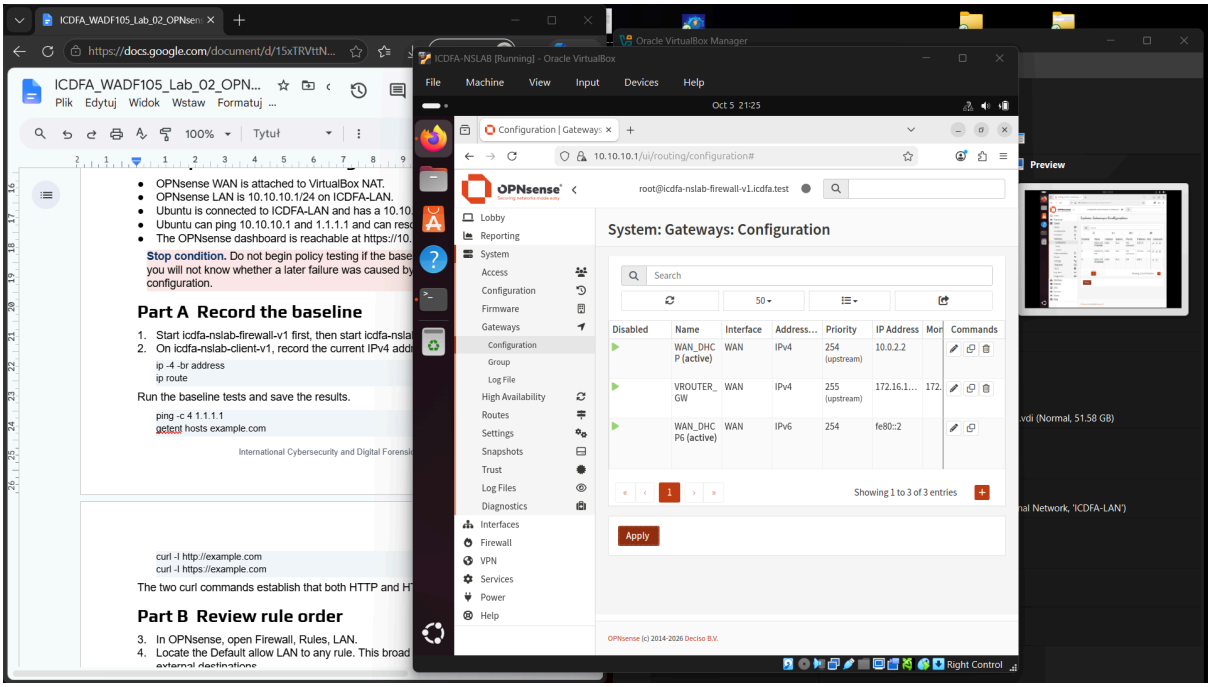




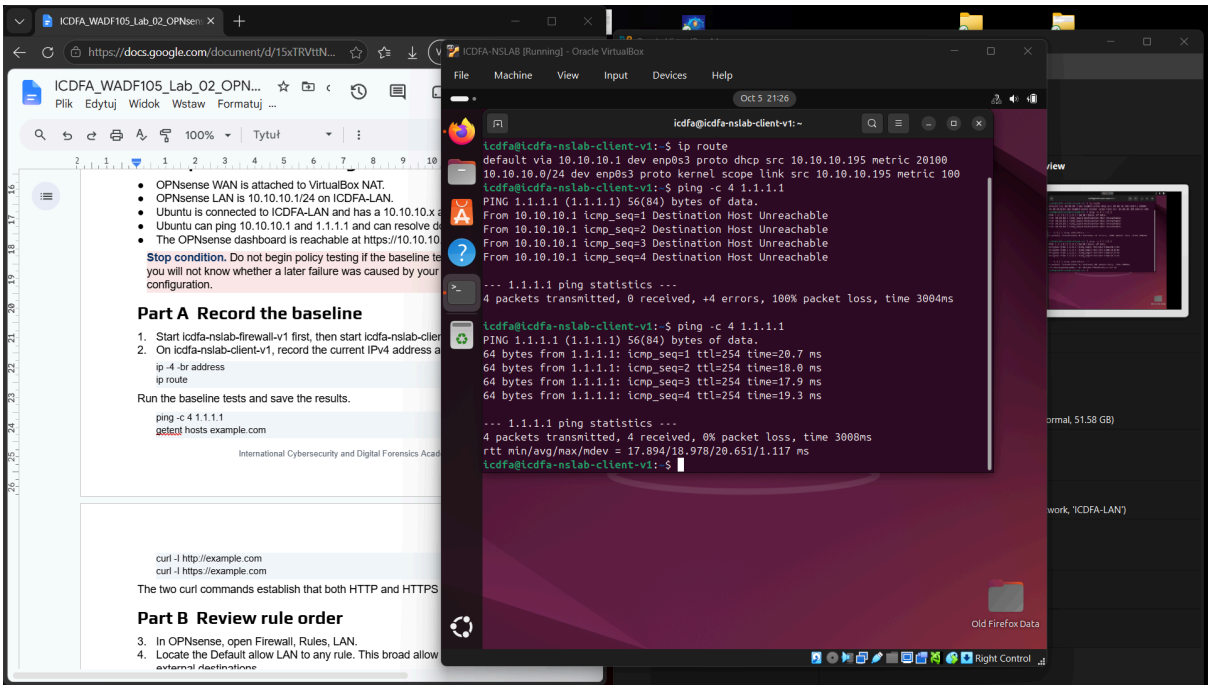
How I Resolved It Permanently:

1. Instead of applying temporary terminal fixes, I logged into the OPNsense web management interface and navigated to **Interfaces > Assignments** to permanently map the correct physical network adapters to **LAN** and **WAN**.
2. I went to **System > Routing > Gateways**, edited the **WAN_DHCP** IPv4 gateway entry, checked the box to enforce it as the **Upstream Gateway**, and saved the changes.

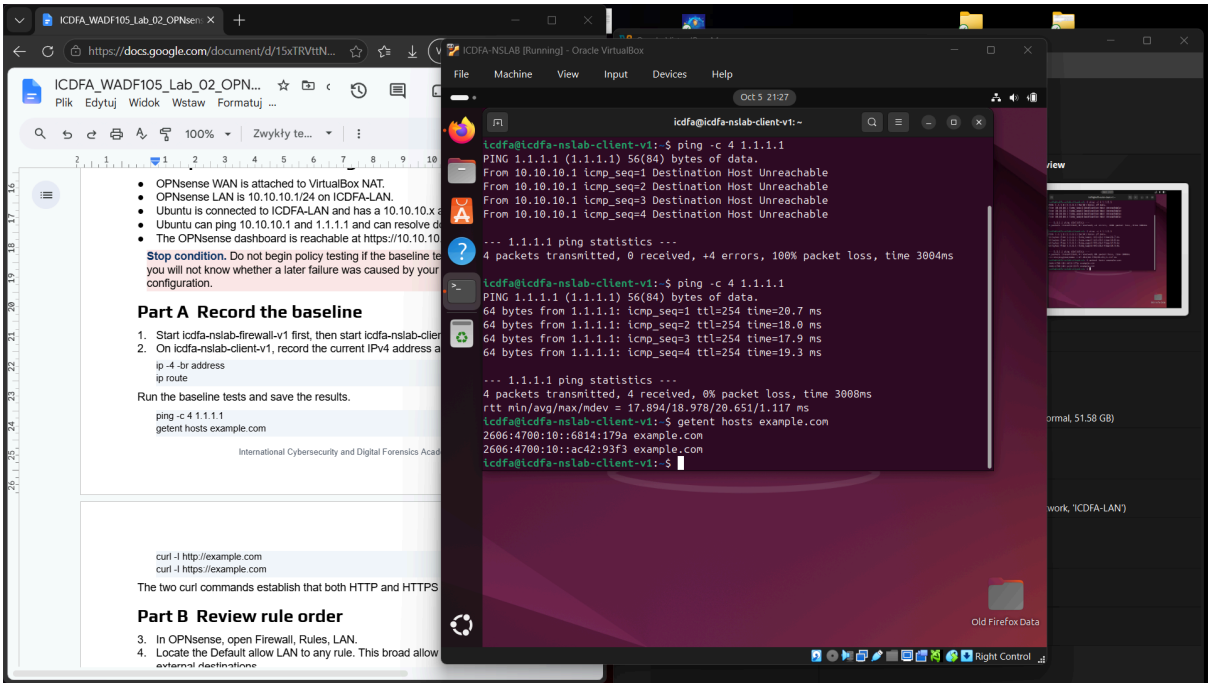
3. Under **Firewall > Rules > LAN**, I verified my interface rules and clicked **Apply changes** so OPNsense wrote the configuration directly to its permanent system storage.



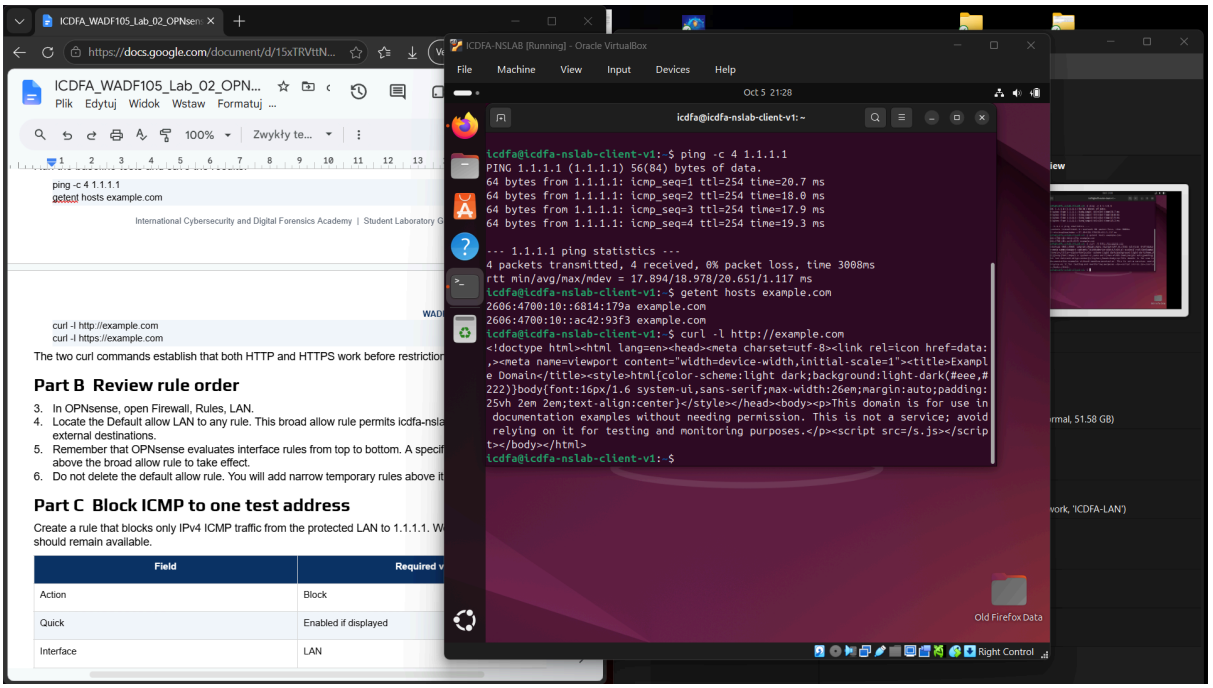
Verification: After applying the permanent configuration in OPNsense and rebooting to test persistence, I returned to my Ubuntu terminal and ran ping -c 4 1.1.1.1. The command succeeded with **0% packet loss**, confirming that the LAN/WAN interface routing was fixed for good.



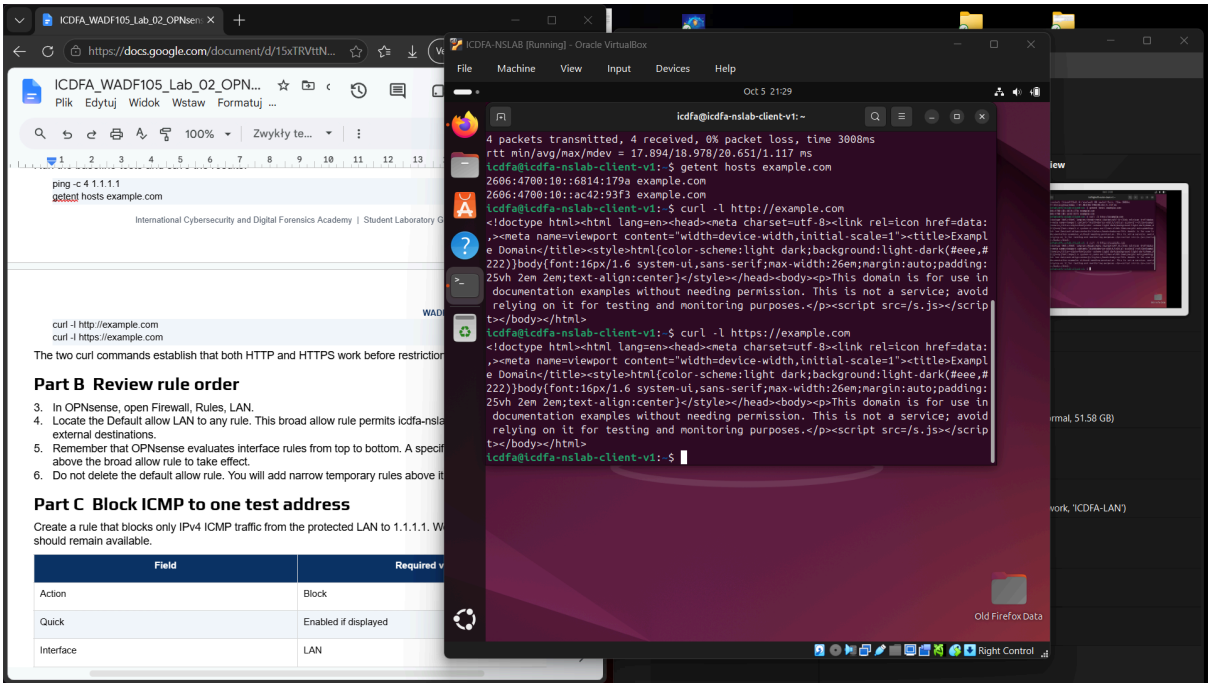
Running command “getent hosts example.com” and results :



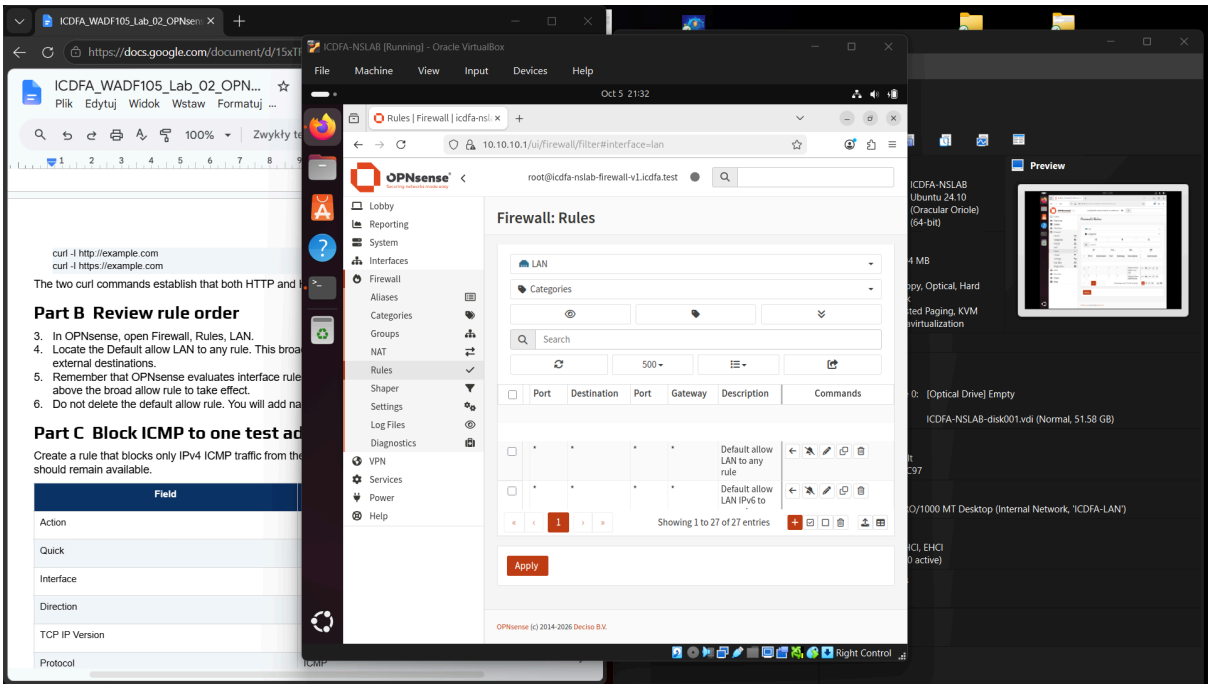
Running command “curl -I http://example.com” and results:

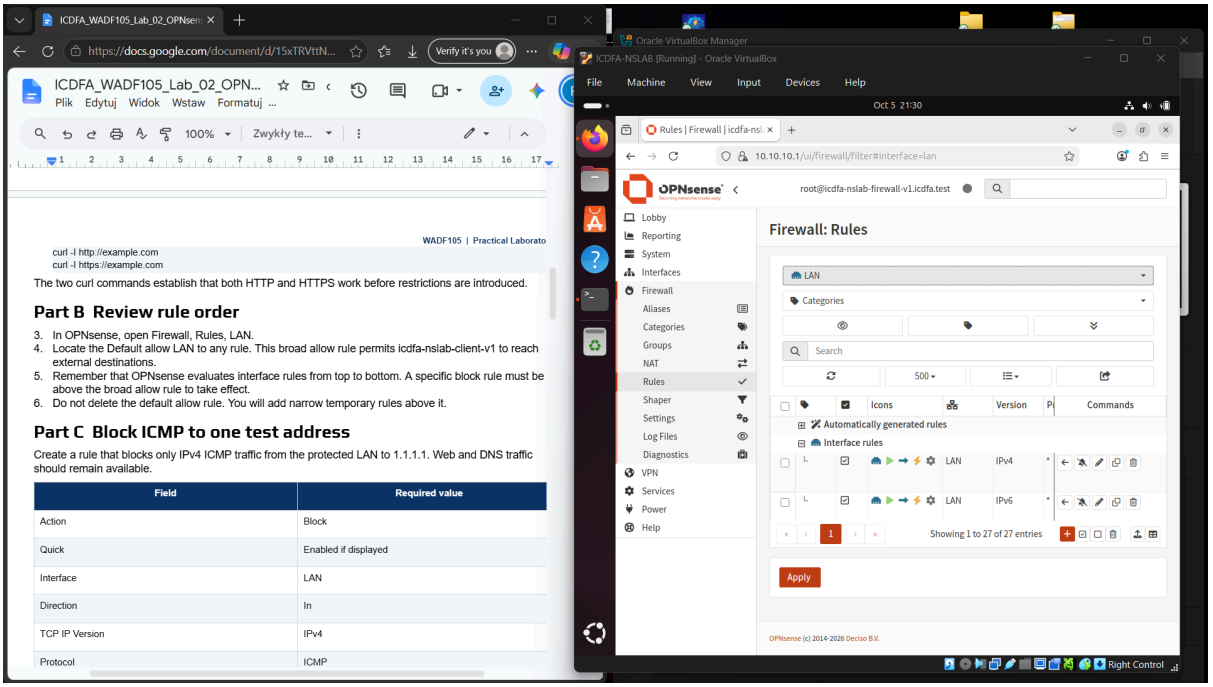


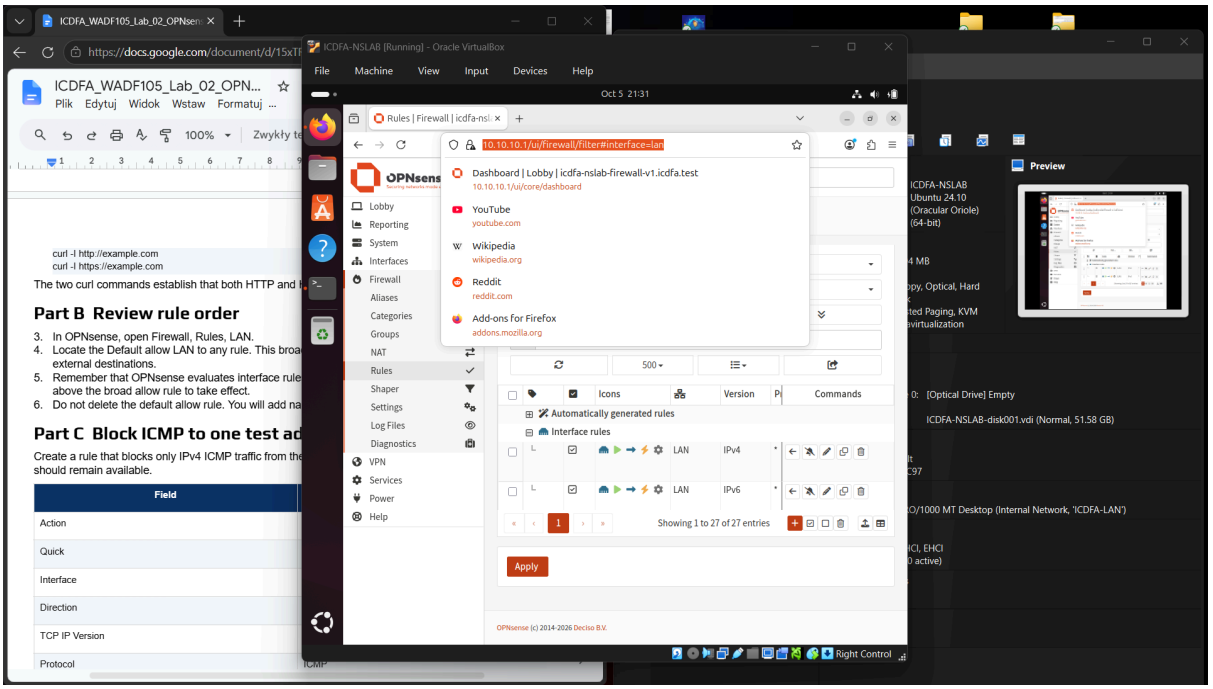
Running command “curl -I https://example.com” and results:



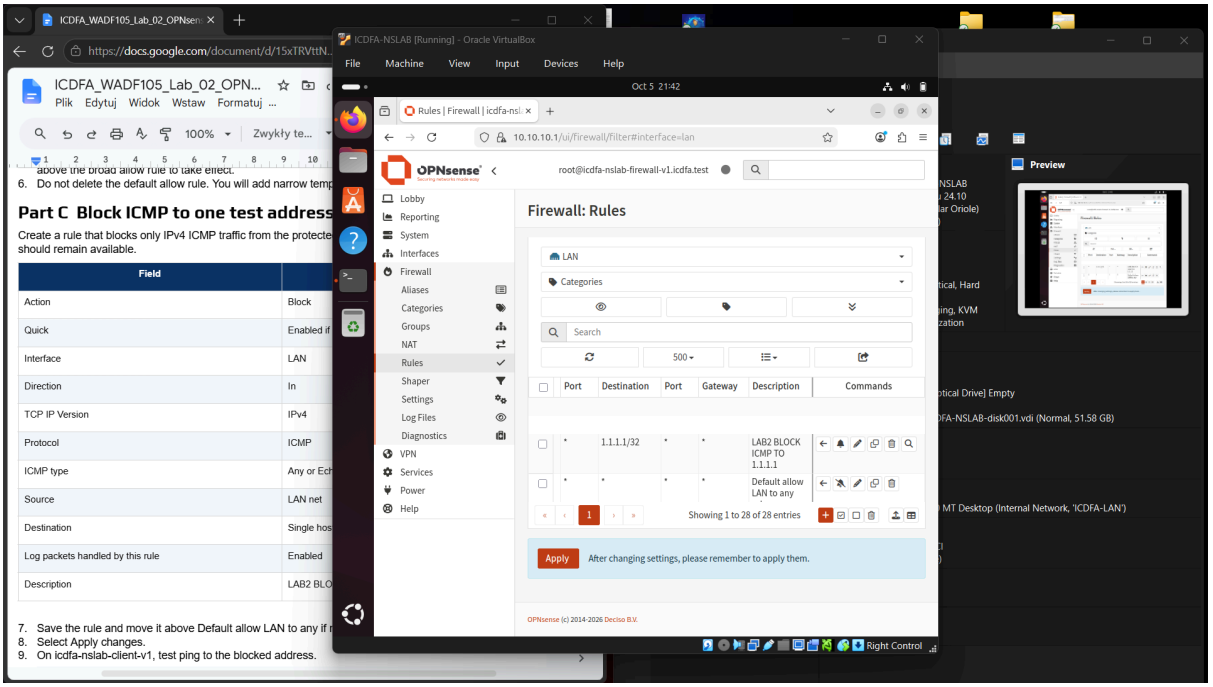
Opening OPNsense and locating LAN:

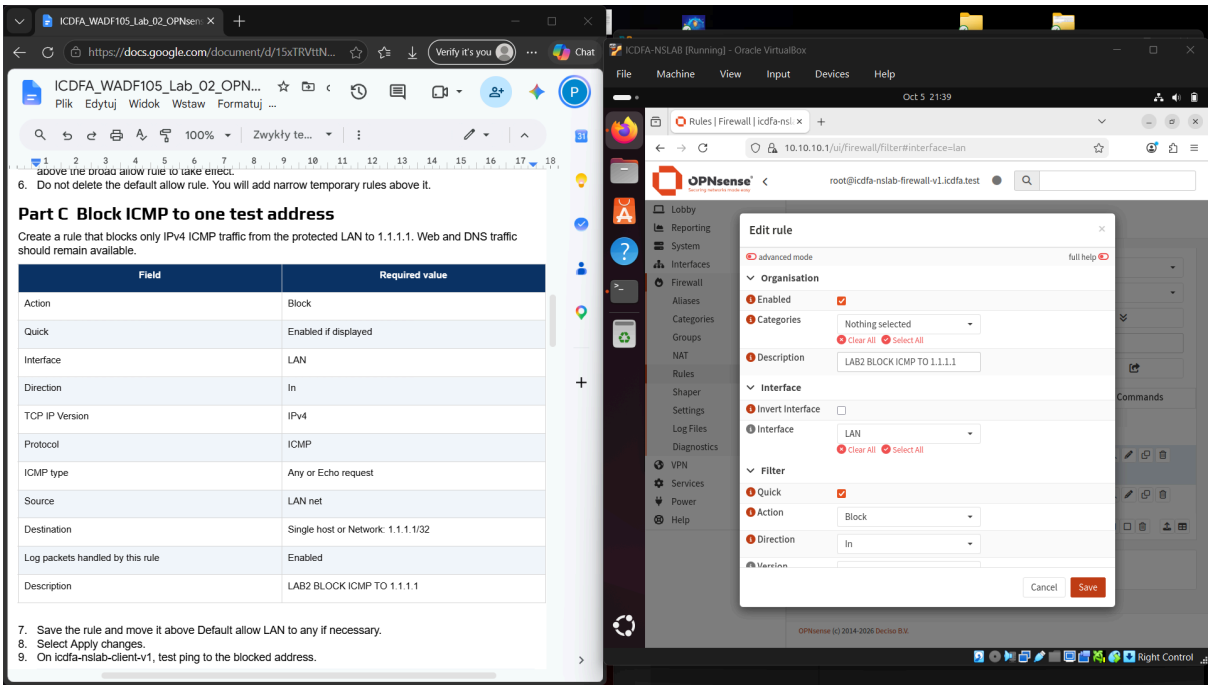




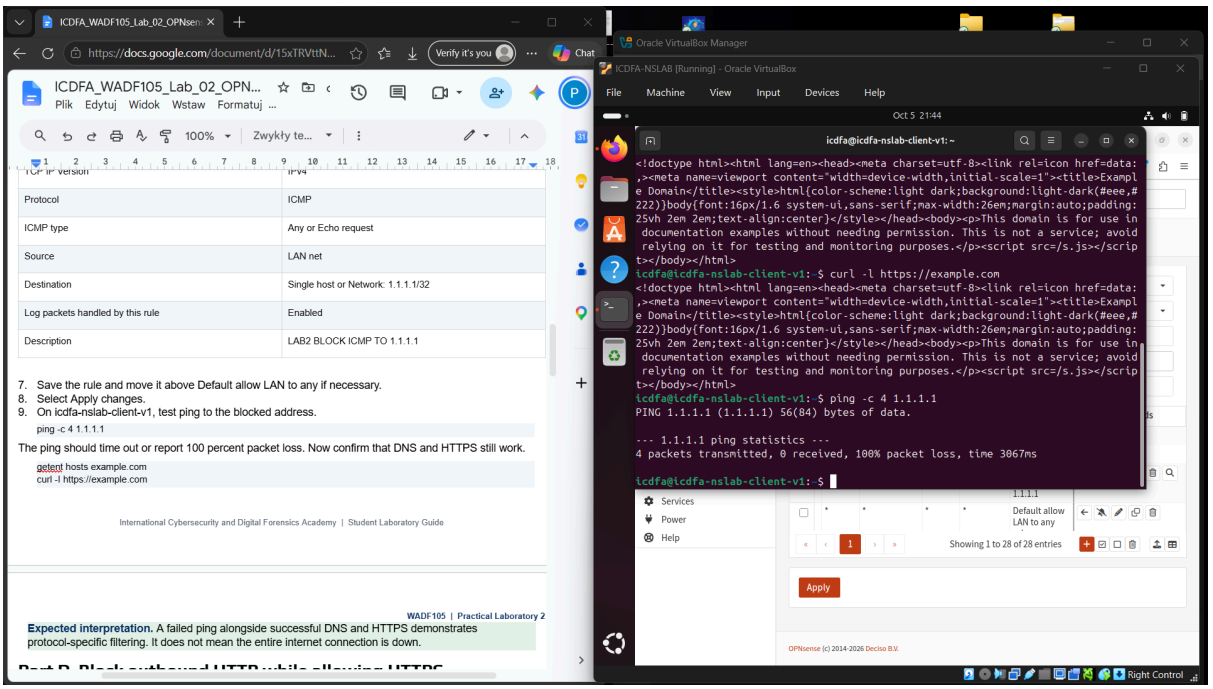


Creating block:

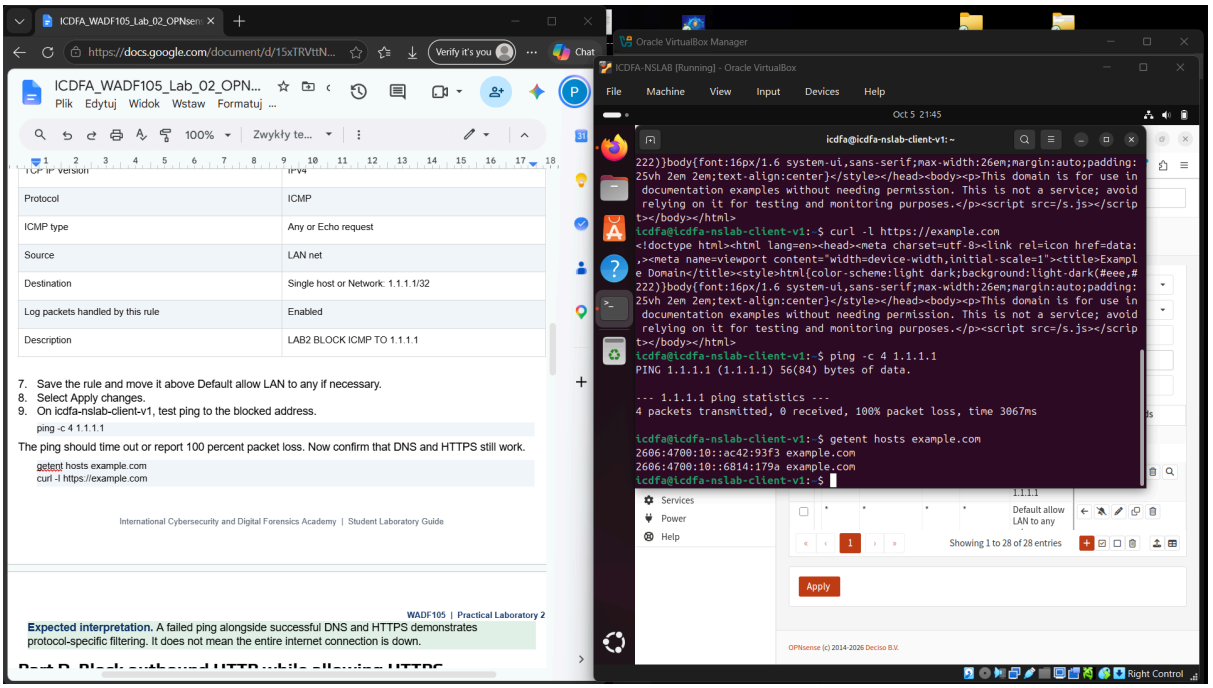




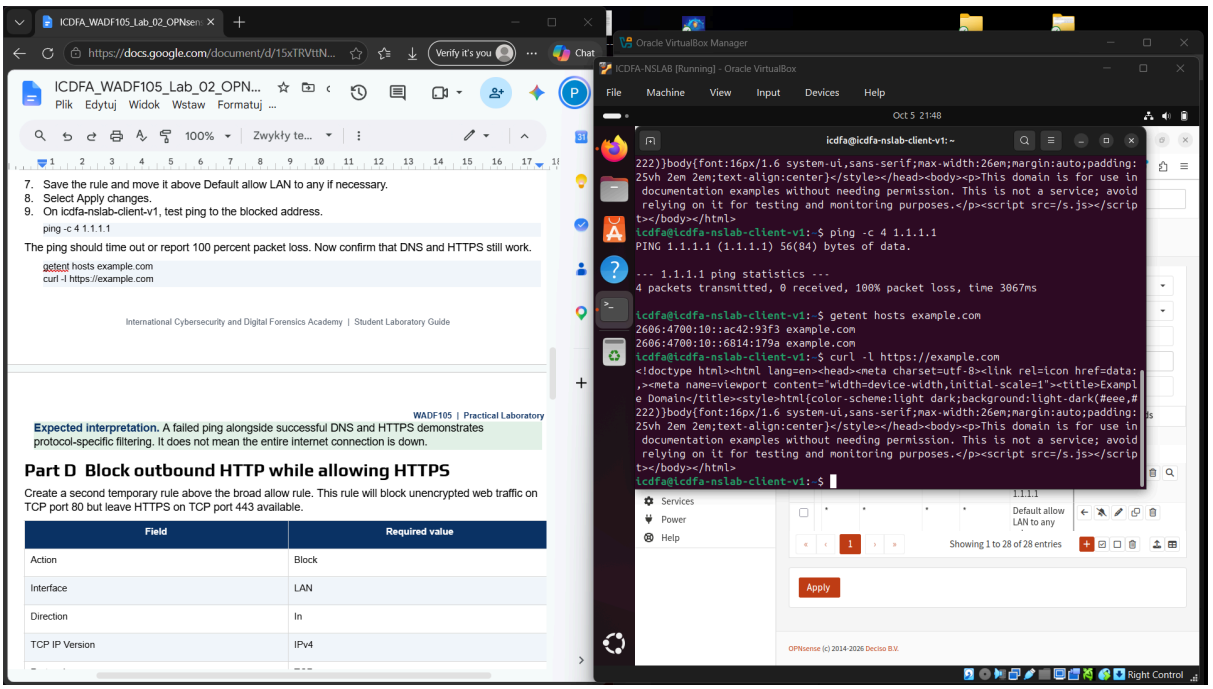
Testing the block on Ubuntu's Terminal running command "ping -c 4 1.1.1.1". Results in successful block and 100% packet loss:



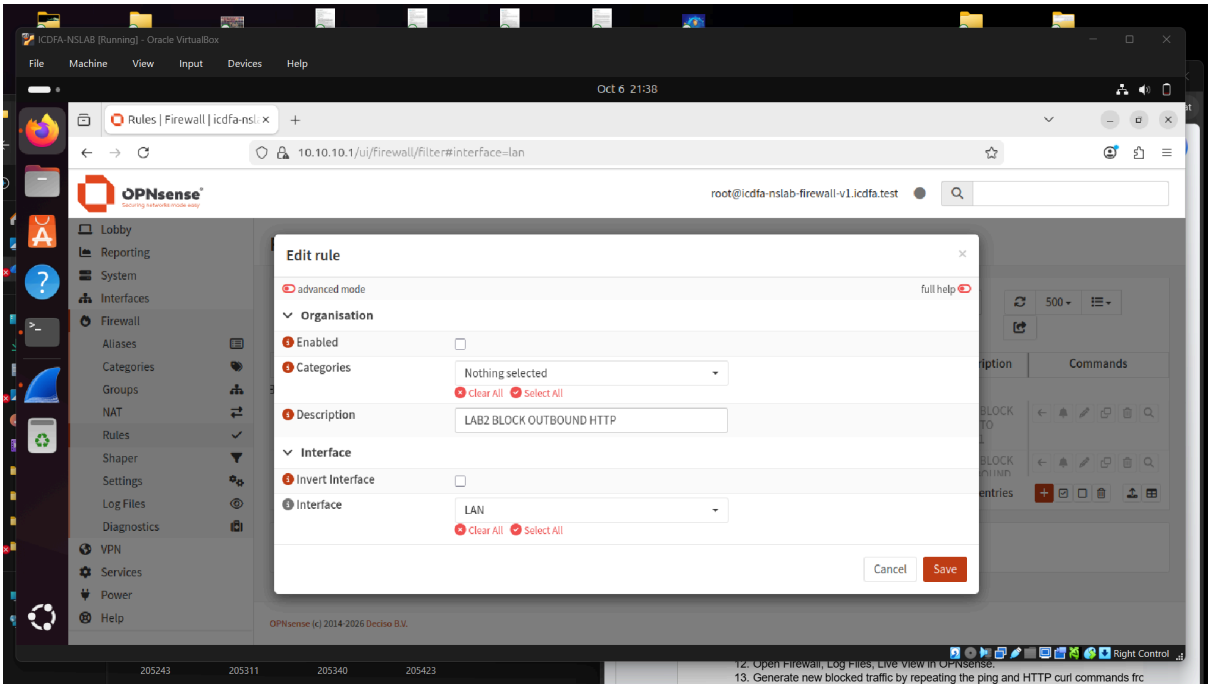
Running command “getent hosts example.com”, recording results:



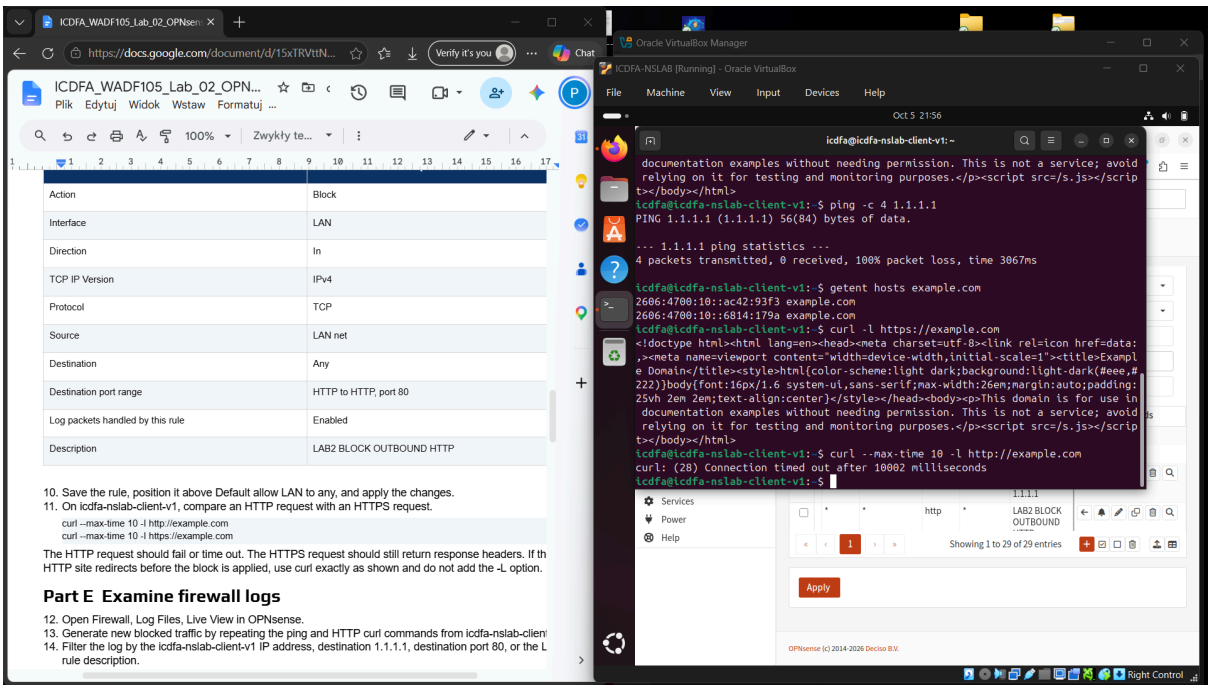
Running command “curl -I https://example.com”, recording results:

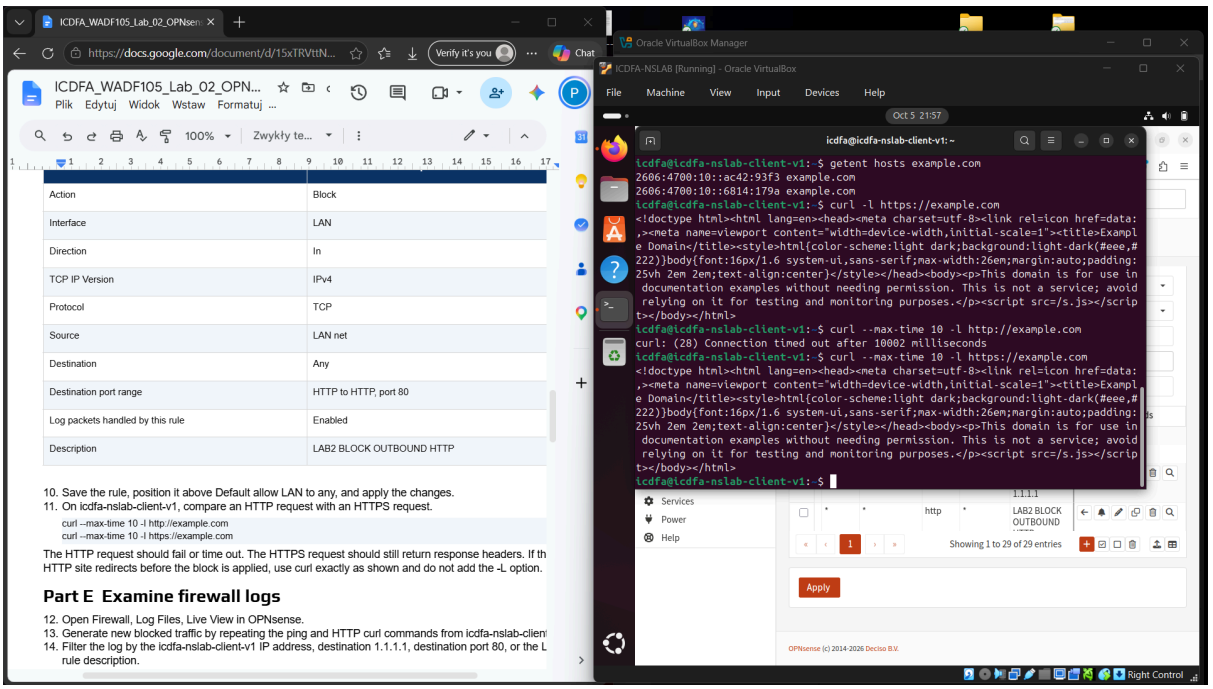


Creating second block:

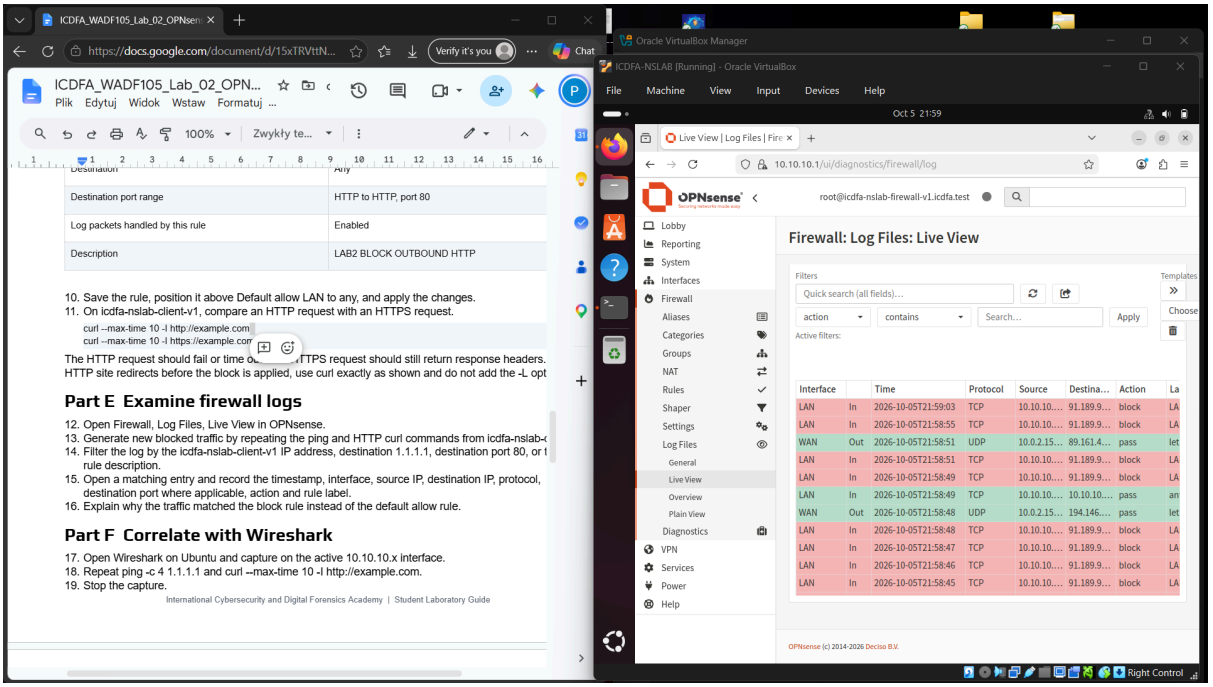


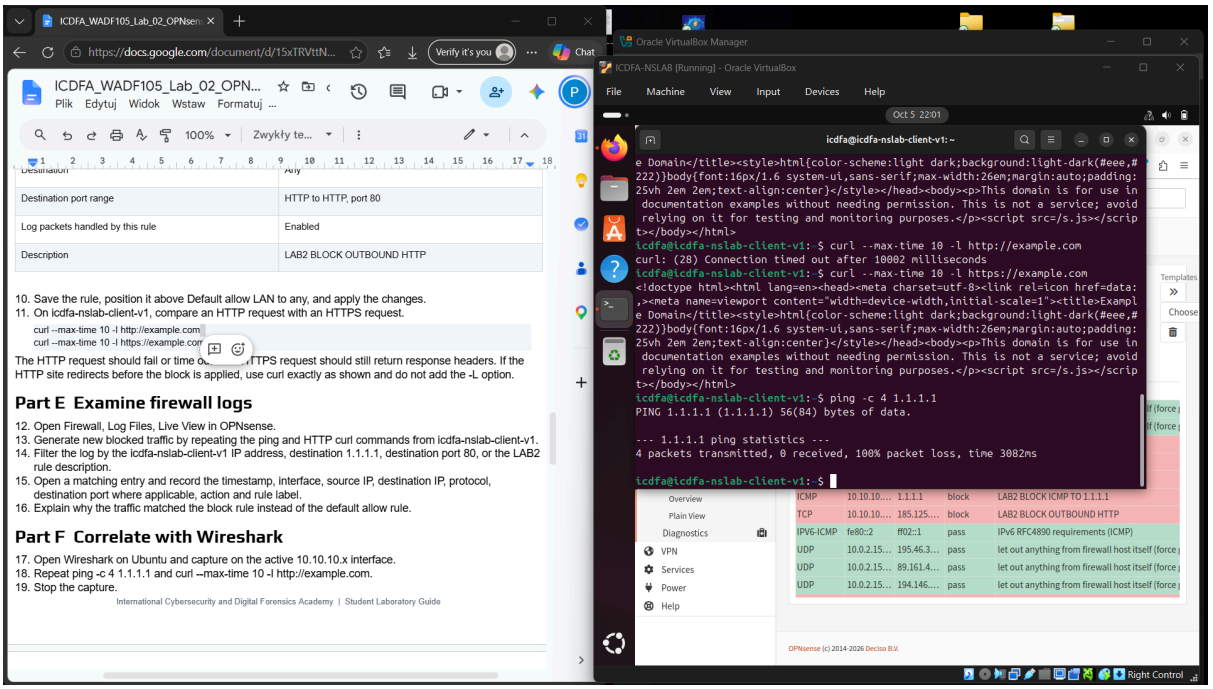
Comparing HTTP request with an HTTPS request, resulting in successful delay in HTTP request and correct return of heathers:

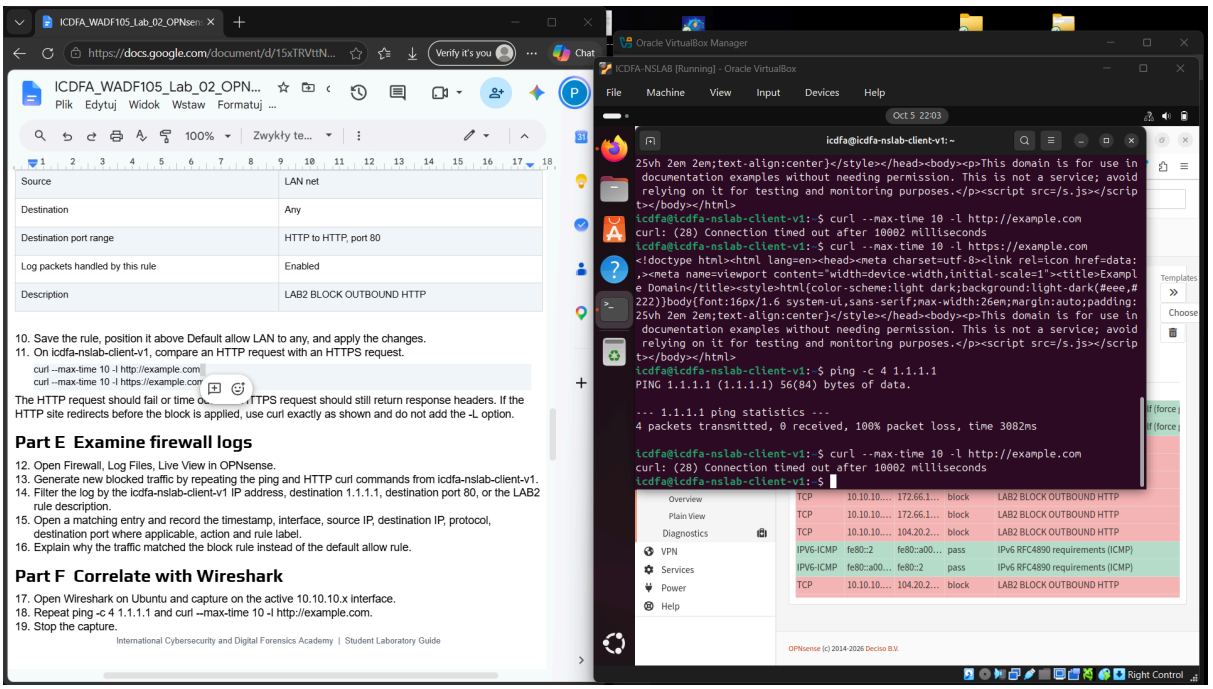




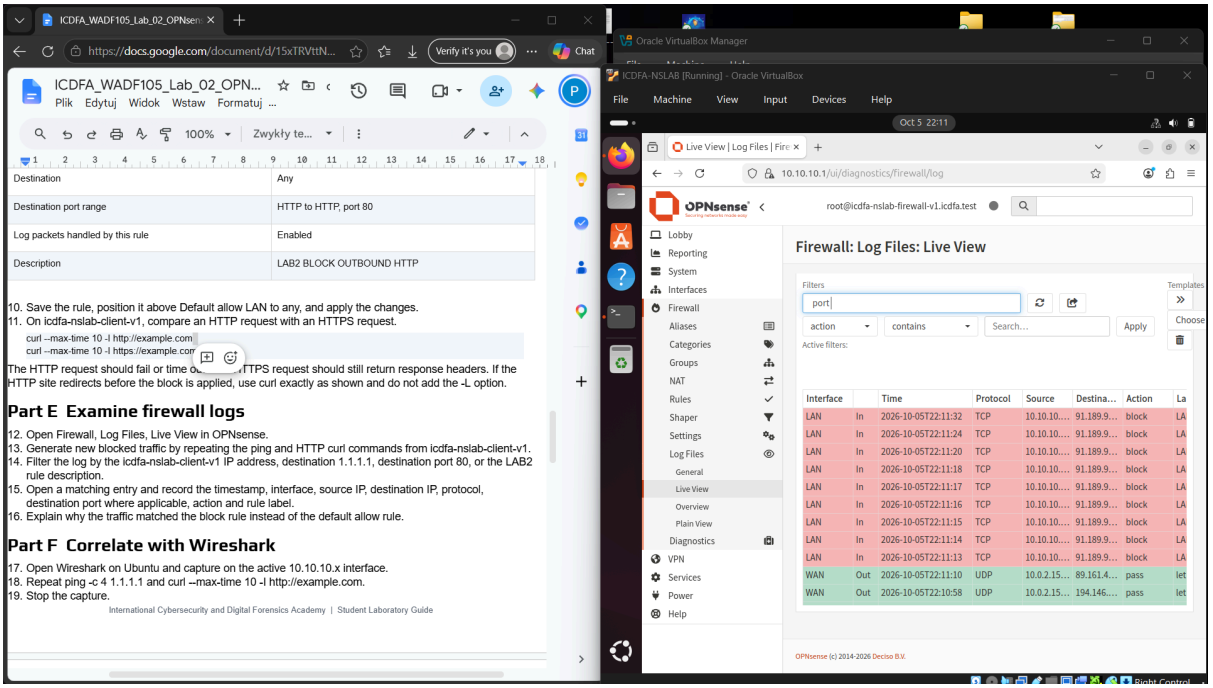
Generating new blocked traffic by repeating the ping and HTTP curl commands from Client:

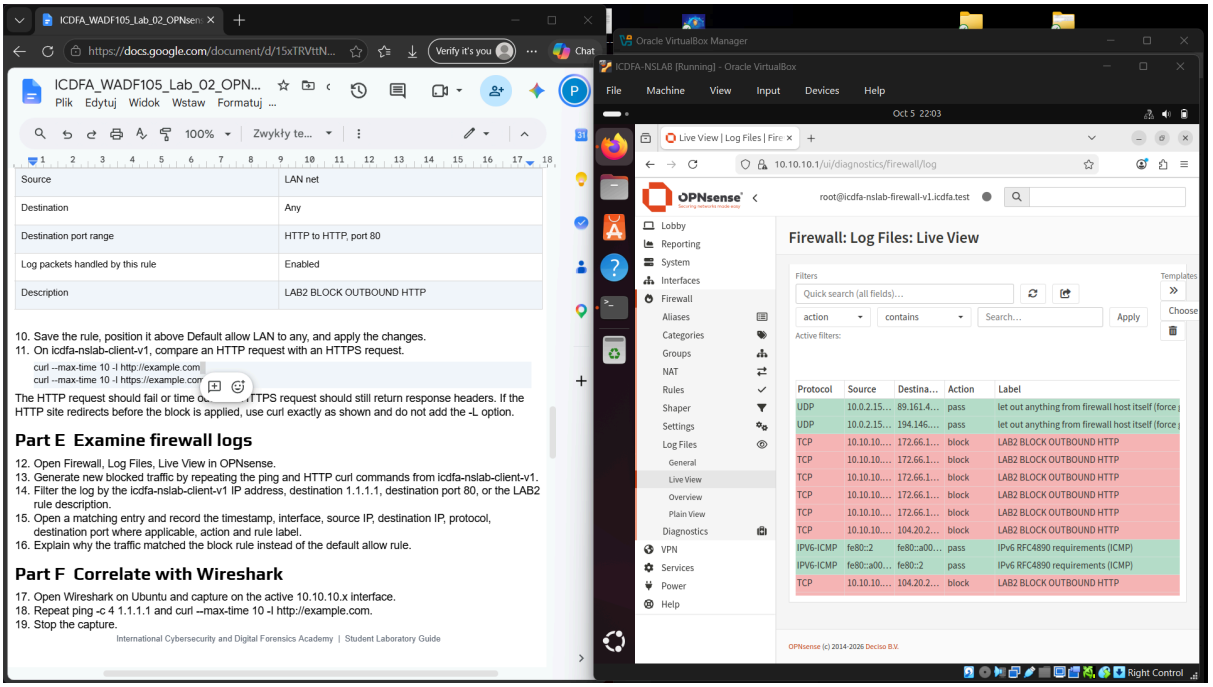

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https://docs.google.com/document/d/15xTRVttN...
Verify it's you
Chat
ICDFA_WADF105_Lab_02_OPN...
Plik Edytuj Widok Wstaw Formatuj ...
Destination port range
HTTP to HTTP, port 80
Log packets handled by this rule
Enabled
Description
LAB2 BLOCK OUTBOUND HTTP
10. Save the rule, position it above Default allow LAN to any, and apply the changes.
11. On icdfa-nslab-client-v1, compare an HTTP request with an HTTPS request.
curl --max-time 10 -I http://example.com
curl --max-time 10 -I https://example.com
Part E Examine firewall logs
12. Open Firewall, Log Files, Live View in OPNsense.
13. Generate new blocked traffic by repeating the ping and HTTP curl commands from icdfa-nslab-client-v1.
14. Filter the log by the icdfa-nslab-client-v1 IP address, destination 1.1.1.1, destination port 80, or the LAB2 rule description.
15. Open a matching entry and record the timestamp, interface, source IP, destination IP, protocol, destination port where applicable, action and rule label.
16. Explain why the traffic matched the block rule instead of the default allow rule.
Part F Correlate with Wireshark
17. Open Wireshark on Ubuntu and capture on the active 10.10.10.x interface.
18. Repeat ping -c 4 1.1.1.1 and curl --max-time 10 -I http://example.com.
19. Stop the capture.
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Oracle VirtualBox Manager
ICDFA-NSLAB [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Live View | Log Files | Fire
10.10.10.1/ui/diagnostics/firewall/log
OPNsense
root@icdfa-nslab-firewall-v1.icdfa.test
Firewall: Log Files: Live View
Lobby
Reporting
System
Interfaces
Firewall
Aliases
Categories
Groups
NAT
Rules
Shaper
Settings
Log Files
General
Live View
Overview
Plain View
Diagnostics
VPN
Services
Power
Help
Interface Time Protocol Source Destina... Action
Right Control

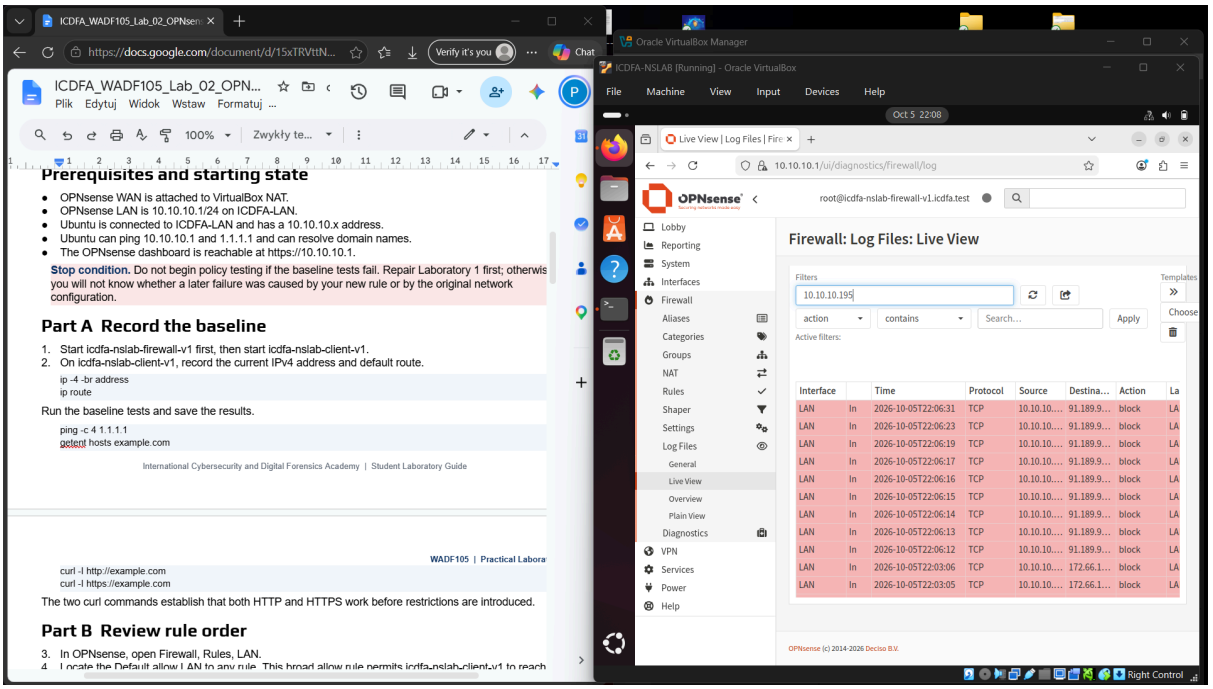

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Verify it's you
Chat
ICDFA_WADF105_Lab_02_OPN...
Plik Edytuj Widok Wstaw Formatuj ...
Destination port range
HTTP to HTTP, port 80
Log packets handled by this rule
Enabled
Description
LAB2 BLOCK OUTBOUND HTTP
10. Save the rule, position it above Default allow LAN to any, and apply the changes.
11. On icdfa-nslab-client-v1, compare an HTTP request with an HTTPS request.
curl --max-time 10 -I http://example.com
curl --max-time 10 -I https://example.com
Part E Examine firewall logs
12. Open Firewall, Log Files, Live View in OPNsense.
13. Generate new blocked traffic by repeating the ping and HTTP curl commands from icdfa-nslab-client-v1.
14. Filter the log by the icdfa-nslab-client-v1 IP address, destination 1.1.1.1, destination port 80, or the LAB2 rule description.
15. Open a matching entry and record the timestamp, interface, source IP, destination IP, protocol, destination port where applicable, action and rule label.
16. Explain why the traffic matched the block rule instead of the default allow rule.
Part F Correlate with Wireshark
17. Open Wireshark on Ubuntu and capture on the active 10.10.10.x interface.
18. Repeat ping -c 4 1.1.1.1 and curl --max-time 10 -I http://example.com.
19. Stop the capture.
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Oracle VirtualBox Manager
ICDFA-NSLAB [Running] - Oracle VirtualBox
File Machine View Input Devices Help
icdfa@icdfa-nslab-client-v1: ~
icdfa@icdfa-nslab-client-v1:~$ curl --max-time 10 -I http://example.com
curl: (28) Connection timed out after 10002 milliseconds
icdfa@icdfa-nslab-client-v1:~$ curl --max-time 10 -I https://example.com
icdfa@icdfa-nslab-client-v1:~$ ping -c 4 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.
--- 1.1.1.1 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3082ms
icdfa@icdfa-nslab-client-v1:~$
Right Control

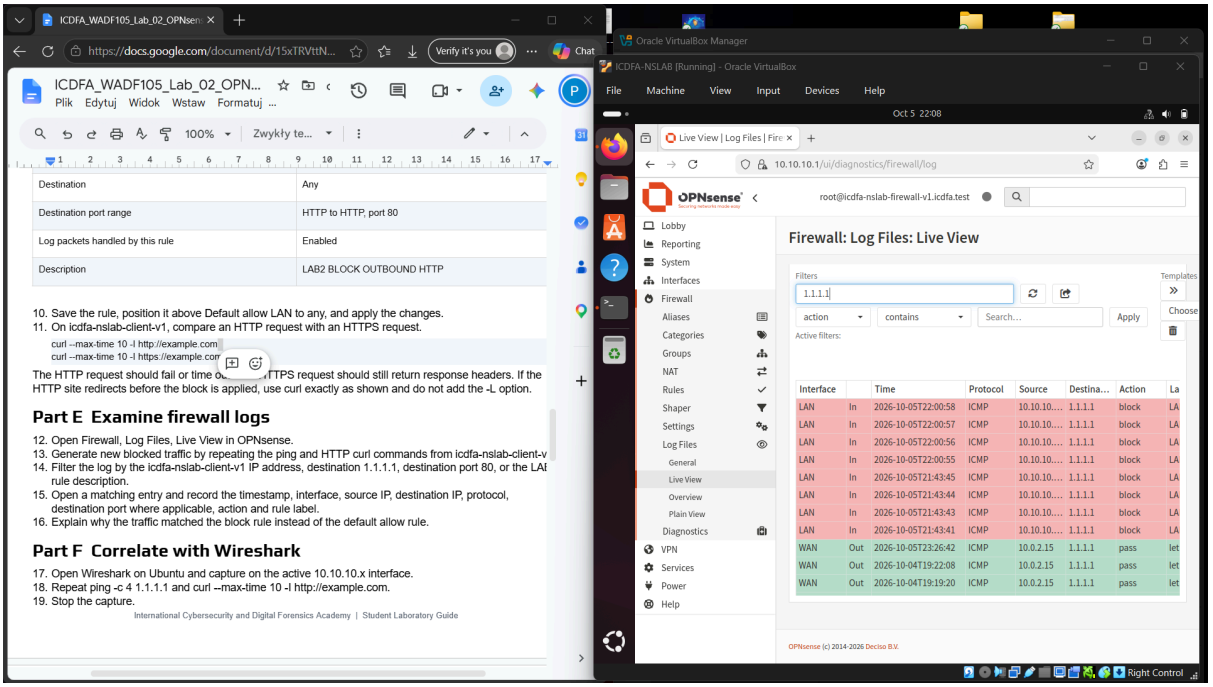


Filtering the log by the icdfa-nslab-client-v1 IP address, destination 1.1.1.1, destination port 80, or the LAB2 rule description:

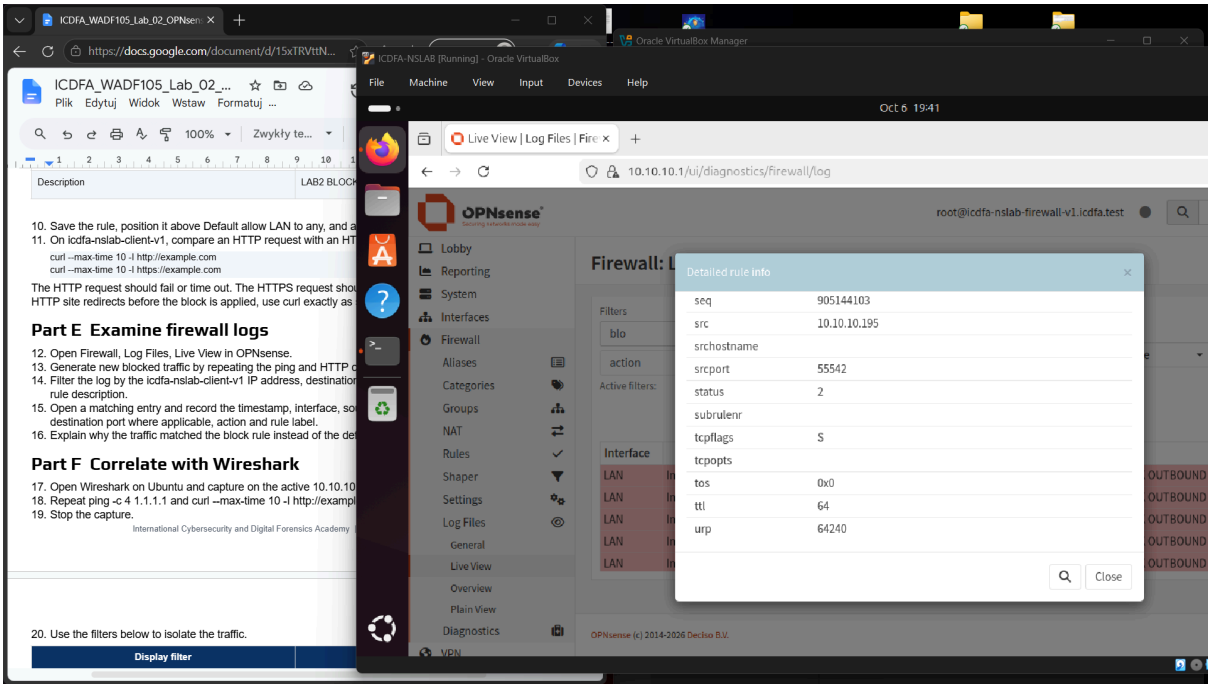


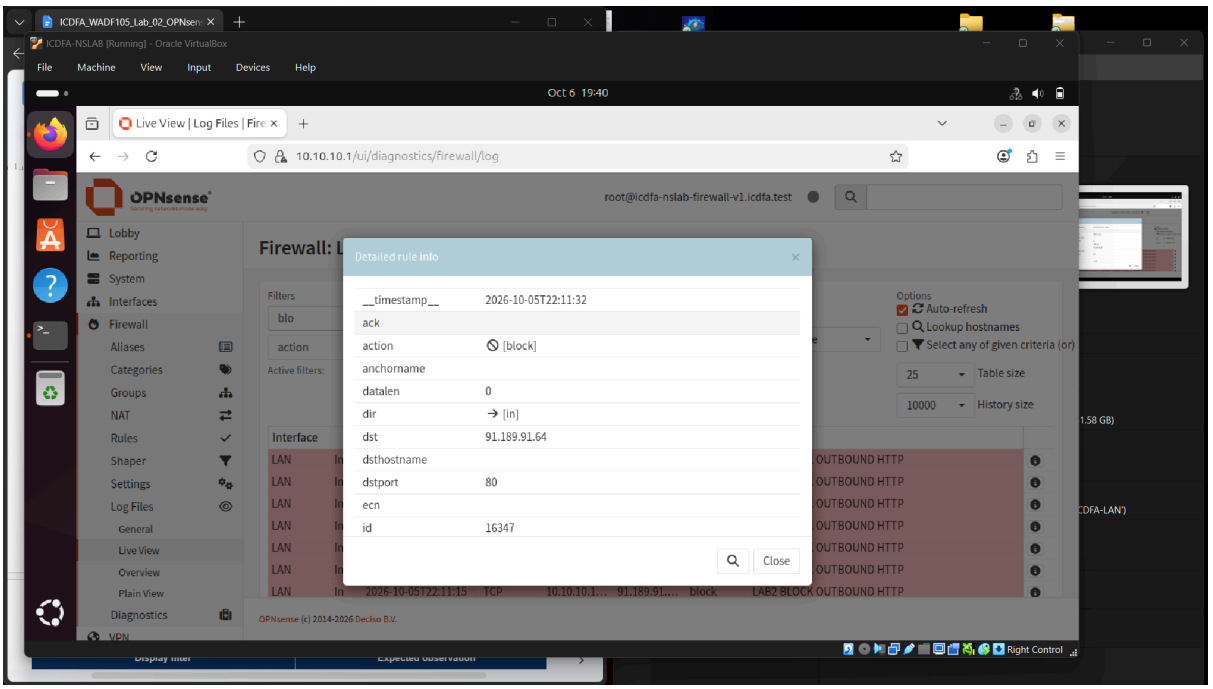

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100%
Zwykły te...
Source
LAN net
Destination
Any
Destination port range
HTTP to HTTP, port 80
Log packets handled by this rule
Enabled
Description
LAB2 BLOCK OUTBOUND HTTP
10. Save the rule, position it above Default allow LAN to any, and apply the changes.
11. On icdfa-nslab-client-v1, compare an HTTP request with an HTTPS request.
curl --max-time 10 -I http://example.com
curl --max-time 10 -I https://example.com
The HTTP request should fail or time o... HTTPS request should still return response headers. If the HTTP site redirects before the block is applied, use curl exactly as shown and do not add the -L option.
Part E Examine firewall logs
12. Open Firewall, Log Files, Live View in OPNsense.
13. Generate new blocked traffic by repeating the ping and HTTP curl commands from icdfa-nslab-client-v1.
14. Filter the log by the icdfa-nslab-client-v1 IP address, destination 1.1.1.1, destination port 80, or the LAB2 rule description.
15. Open a matching entry and record the timestamp, interface, source IP, destination IP, protocol, destination port where applicable, action and rule label.
16. Explain why the traffic matched the block rule instead of the default allow rule.
Part F Correlate with Wireshark
17. Open Wireshark on Ubuntu and capture on the active 10.10.10.x interface.
18. Repeat ping -c 4 1.1.1.1 and curl --max-time 10 -I http://example.com.
19. Stop the capture.
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Oracle VirtualBox Manager
ICDFA-NSLAB [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Oct 5 22:03
Live View | Log Files | Fire
10.10.10.1/ui/diagnostics/firewall/log
OPNsense
root@icdfa-nslab-firewall-v1.icdfa.test
Lobby
Reporting
System
Interfaces
Firewall
Aliases
Categories
Groups
NAT
Rules
Shaper
Settings
Log Files
General
Live View
Overview
Plain View
Diagnostics
VPN
Services
Power
Help
Firewall: Log Files: Live View
Filters
Quick search (all fields)...
action
contains
Search...
Apply
Active filters:
Templates
Protocol Source Destina... Action Label
UDP 10.0.2.15... 89.161.4... pass let out anything from firewall host itself (force
UDP 10.0.2.15... 194.146.... pass let out anything from firewall host itself (force
TCP 10.10.10.... 172.66.1... block LAB2 BLOCK OUTBOUND HTTP
TCP 10.10.10.... 172.66.1... block LAB2 BLOCK OUTBOUND HTTP
TCP 10.10.10.... 172.66.1... block LAB2 BLOCK OUTBOUND HTTP
TCP 10.10.10.... 172.66.1... block LAB2 BLOCK OUTBOUND HTTP
TCP 10.10.10.... 172.66.1... block LAB2 BLOCK OUTBOUND HTTP
TCP 10.10.10.... 104.20.2... block LAB2 BLOCK OUTBOUND HTTP
IPV6-ICMP fe80::2 fe80::a00... pass IPv6 RFC4890 requirements (ICMP)
IPV6-ICMP fe80::a00... fe80::2 pass IPv6 RFC4890 requirements (ICMP)
TCP 10.10.10.... 104.20.2... block LAB2 BLOCK OUTBOUND HTTP
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Right Control


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Plik Edytuj Widok Wstaw Formatuj ...
100%
Zwykły te...
Prerequisites and starting state
OPNsense WAN is attached to VirtualBox NAT.
OPNsense LAN is 10.10.10.1/24 on ICDFA-LAN.
Ubuntu is connected to ICDFA-LAN and has a 10.10.10.x address.
Ubuntu can ping 10.10.10.1 and 1.1.1.1 and can resolve domain names.
The OPNsense dashboard is reachable at https://10.10.10.1.
Stop condition. Do not begin policy testing if the baseline tests fail. Repair Laboratory 1 first; otherwis
you will not know whether a later failure was caused by your new rule or by the original network configuration.
Part A Record the baseline
1. Start icdfa-nslab-firewall-v1 first, then start icdfa-nslab-client-v1.
2. On icdfa-nslab-client-v1, record the current IPv4 address and default route.
ip -4 -br address
ip route
Run the baseline tests and save the results.
ping -c 4 1.1.1.1
getent hosts example.com
International Cybersecurity and Digital Forensics Academy | Student Laboratory Guide
WADF105 | Practical Labora
curl -I http://example.com
curl -I https://example.com
The two curl commands establish that both HTTP and HTTPS work before restrictions are introduced.
Part B Review rule order
3. In OPNsense, open Firewall, Rules, LAN.
4. Locate the Default allow LAN to any rule. This broad allow rule permits icdfa-nslab-client-v1 to reach
Oracle VirtualBox Manager
ICDFA-NSLAB [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Oct 5 22:08
Live View | Log Files | Fire
10.10.10.1/ui/diagnostics/firewall/log
OPNsense
root@icdfa-nslab-firewall-v1.icdfa.test
Lobby
Reporting
System
Interfaces
Firewall
Aliases
Categories
Groups
NAT
Rules
Shaper
Settings
Log Files
General
Live View
Overview
Plain View
Diagnostics
VPN
Services
Power
Help
Firewall: Log Files: Live View
Filters
10.10.10.195
action
contains
Search...
Apply
Active filters:
Templates
Interface Time Protocol Source Destina... Action La
LAN In 2026-10-05T22:06:31 TCP 10.10.10.... 91.189.9... block LA
LAN In 2026-10-05T22:06:23 TCP 10.10.10.... 91.189.9... block LA
LAN In 2026-10-05T22:06:19 TCP 10.10.10.... 91.189.9... block LA
LAN In 2026-10-05T22:06:17 TCP 10.10.10.... 91.189.9... block LA
LAN In 2026-10-05T22:06:16 TCP 10.10.10.... 91.189.9... block LA
LAN In 2026-10-05T22:06:15 TCP 10.10.10.... 91.189.9... block LA
LAN In 2026-10-05T22:06:14 TCP 10.10.10.... 91.189.9... block LA
LAN In 2026-10-05T22:06:13 TCP 10.10.10.... 91.189.9... block LA
LAN In 2026-10-05T22:06:12 TCP 10.10.10.... 91.189.9... block LA
LAN In 2026-10-05T22:03:06 TCP 10.10.10.... 172.66.1... block LA
LAN In 2026-10-05T22:03:05 TCP 10.10.10.... 172.66.1... block LA
OPNsense (c) 2014-2026 Deciso B.V.
Right Control



Opening a matching entry and recording the timestamp, interface, source IP, destination IP, protocol, destination port where applicable, action and ruling label:



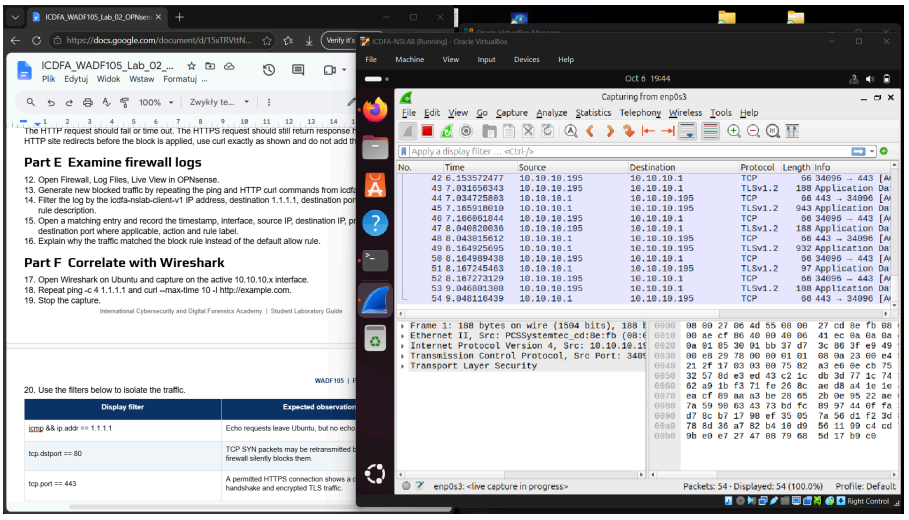


The traffic matched the block rule instead of the default allow rule because OPNsense evaluates firewall rules top-to-bottom in sequential order.

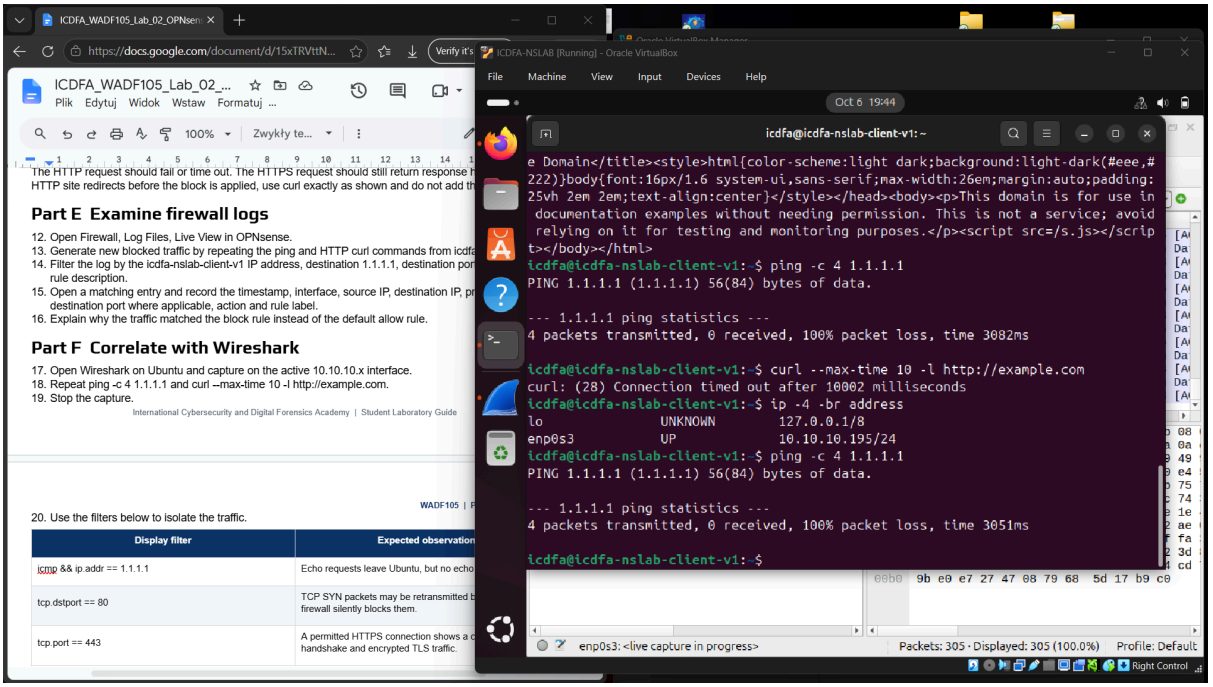
When I generated the test traffic, the packet arrived at the LAN interface and OPNsense started reading the rule list from top to bottom. Since I had placed the specific block rule **above** the `Default allow LAN to any` rule, the packet matched the criteria of the block rule first.

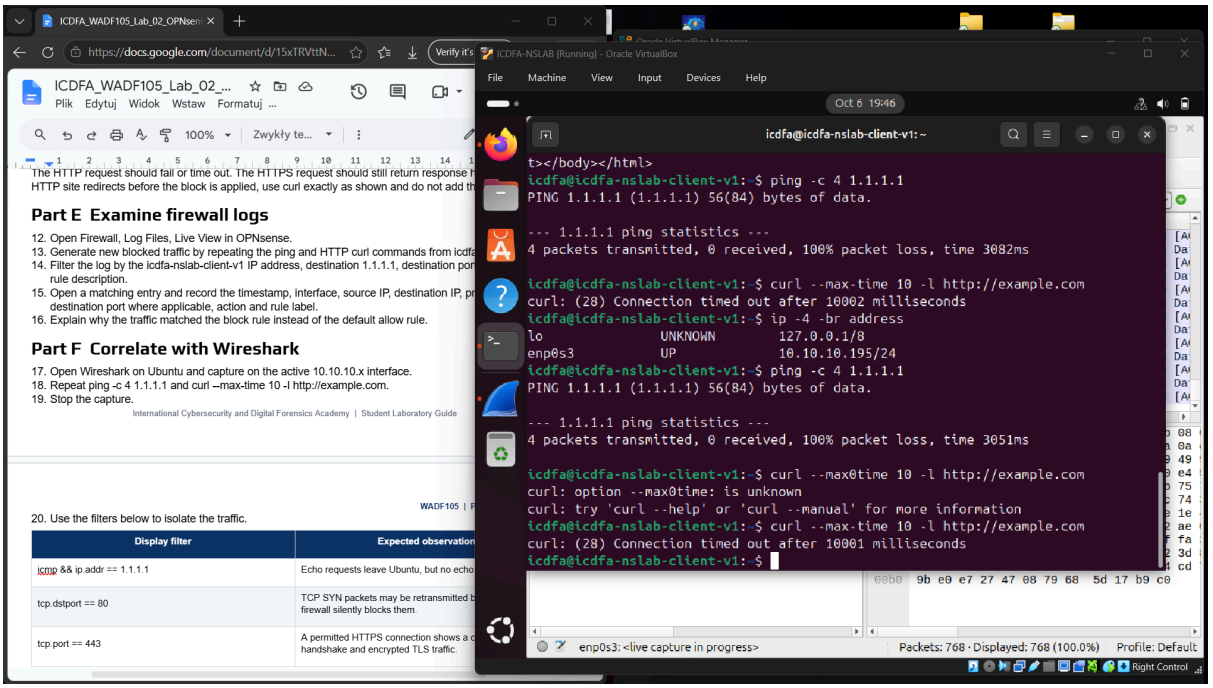
As soon as a match occurs, OPNsense immediately enforces the rule action (dropping or blocking the traffic) and stops evaluating any further rules in the list. Because of this "first-match" behavior, the packet was blocked right away and never made it down to the default allow rule.

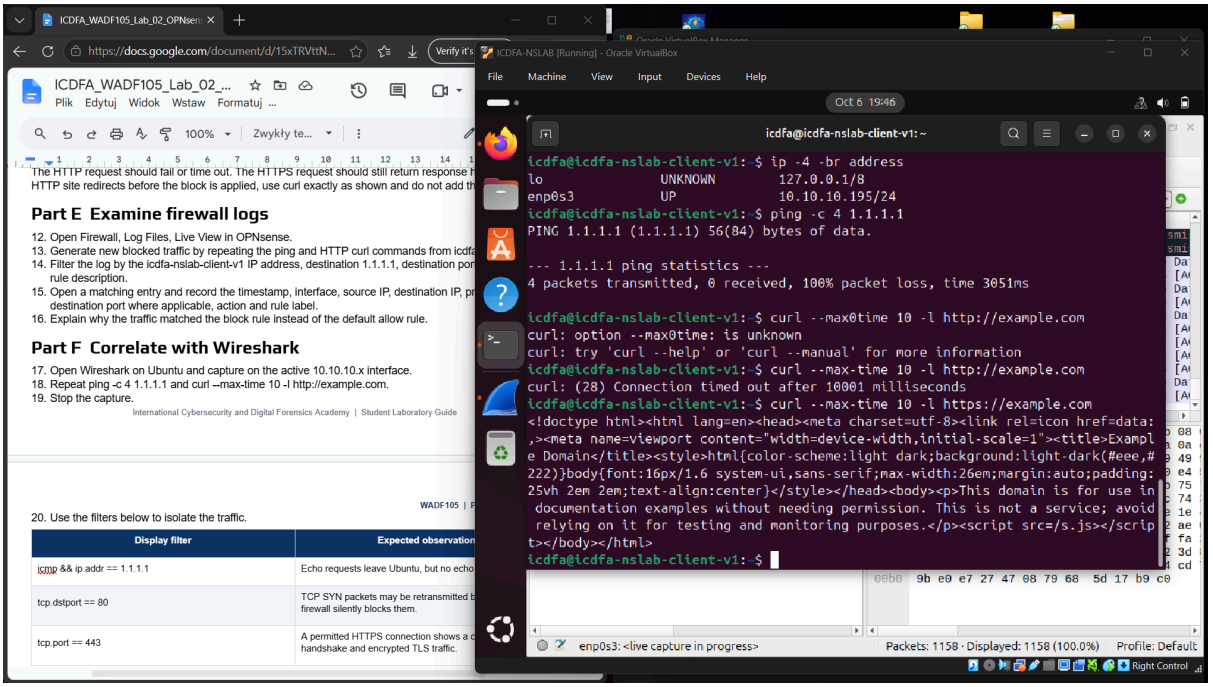
Opening Wireshark and capturing active interface:



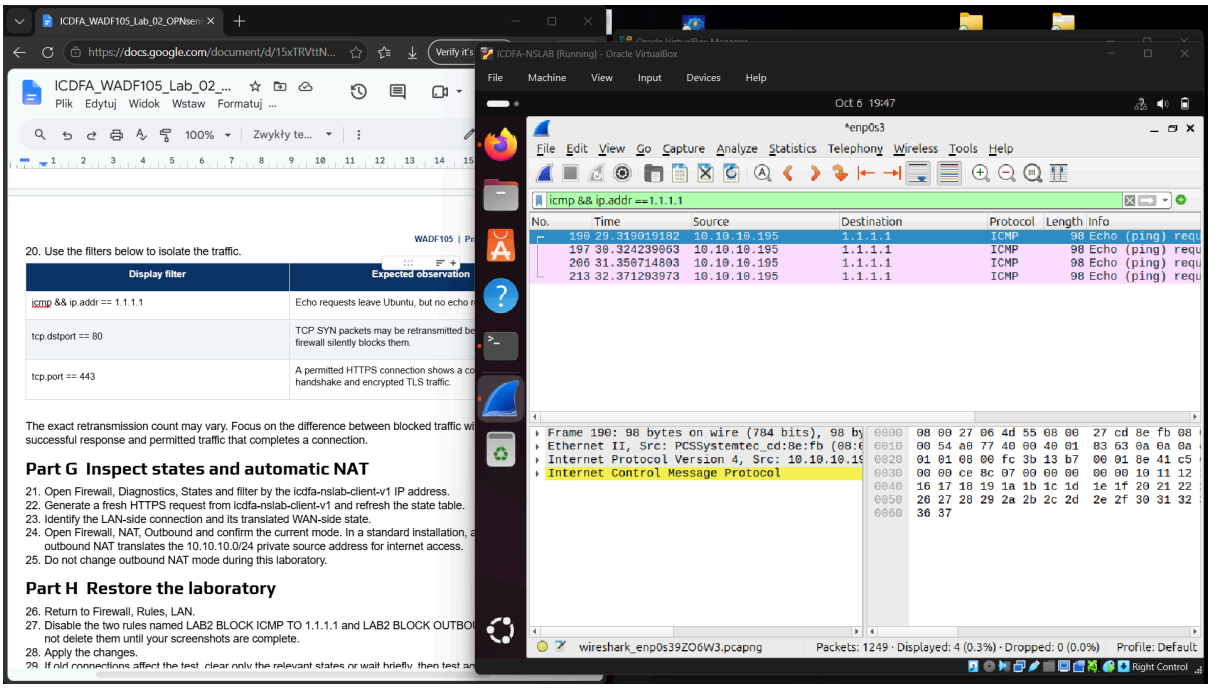
Repeating ping -c 4 1.1.1.1 and curl --max-time 10 -I http://example.com:

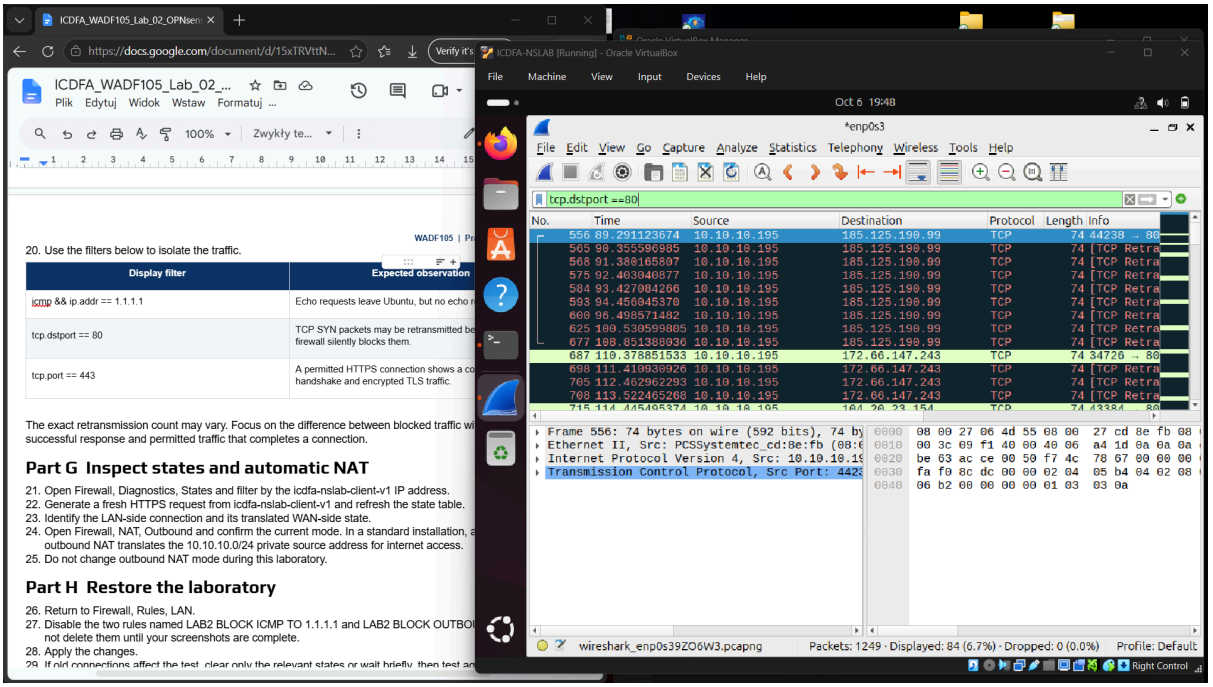


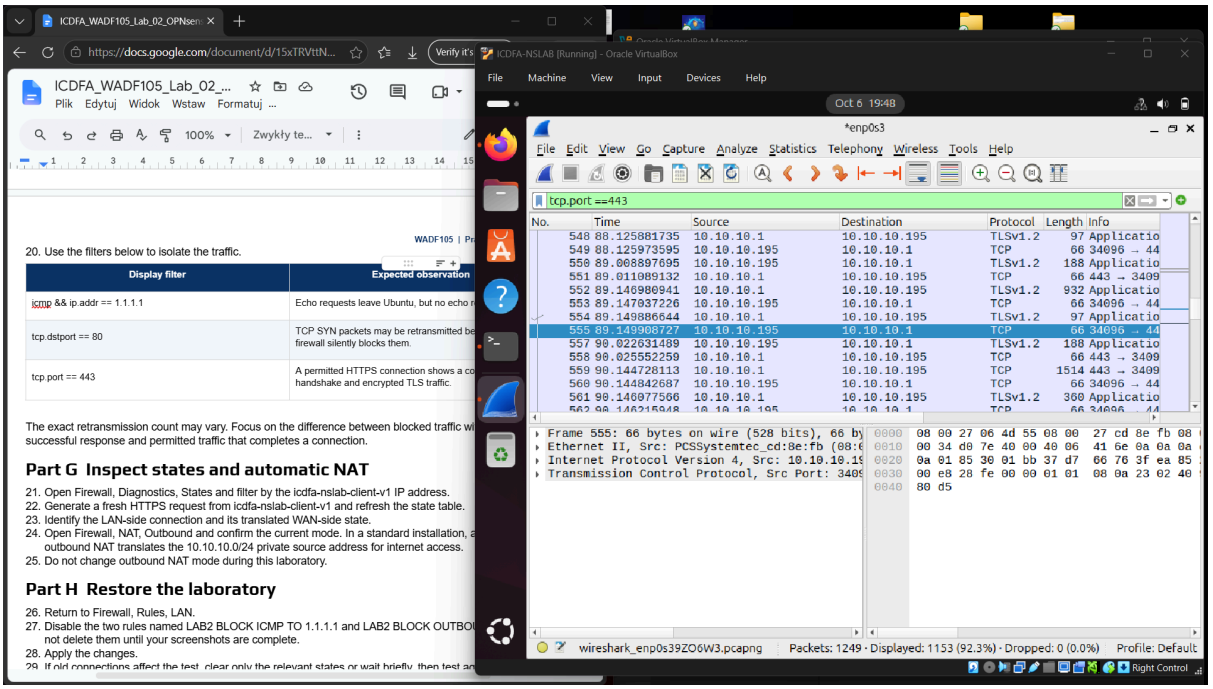




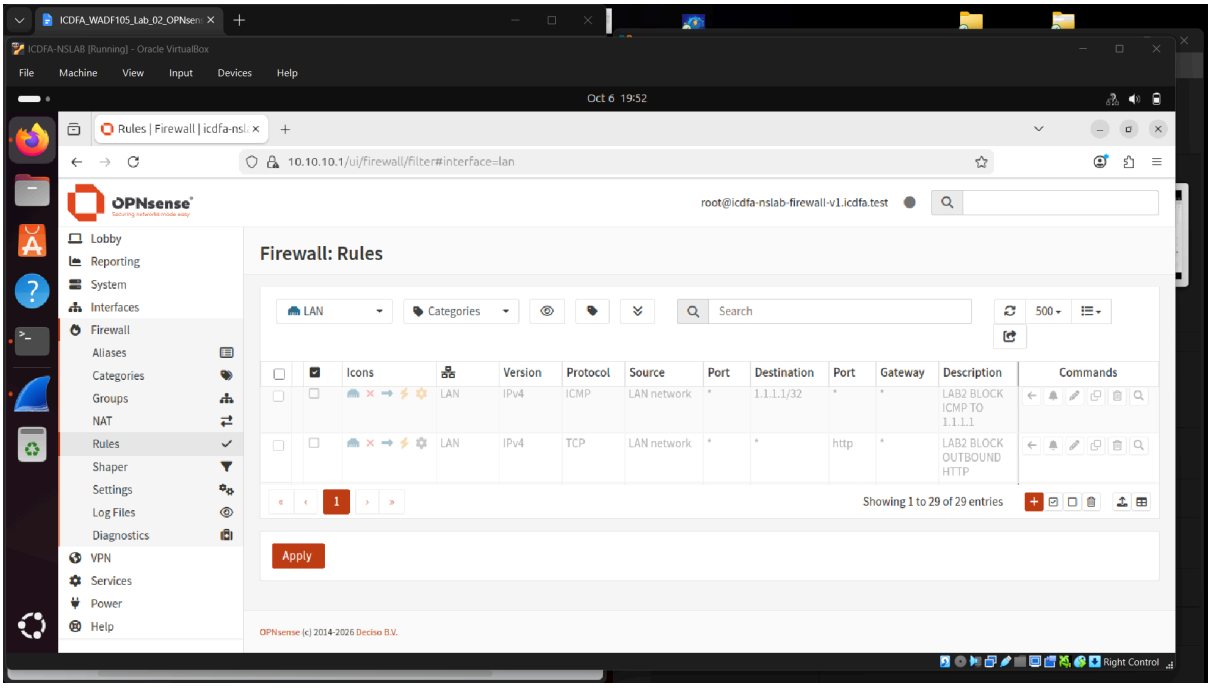
Using filters to isolate the traffic:



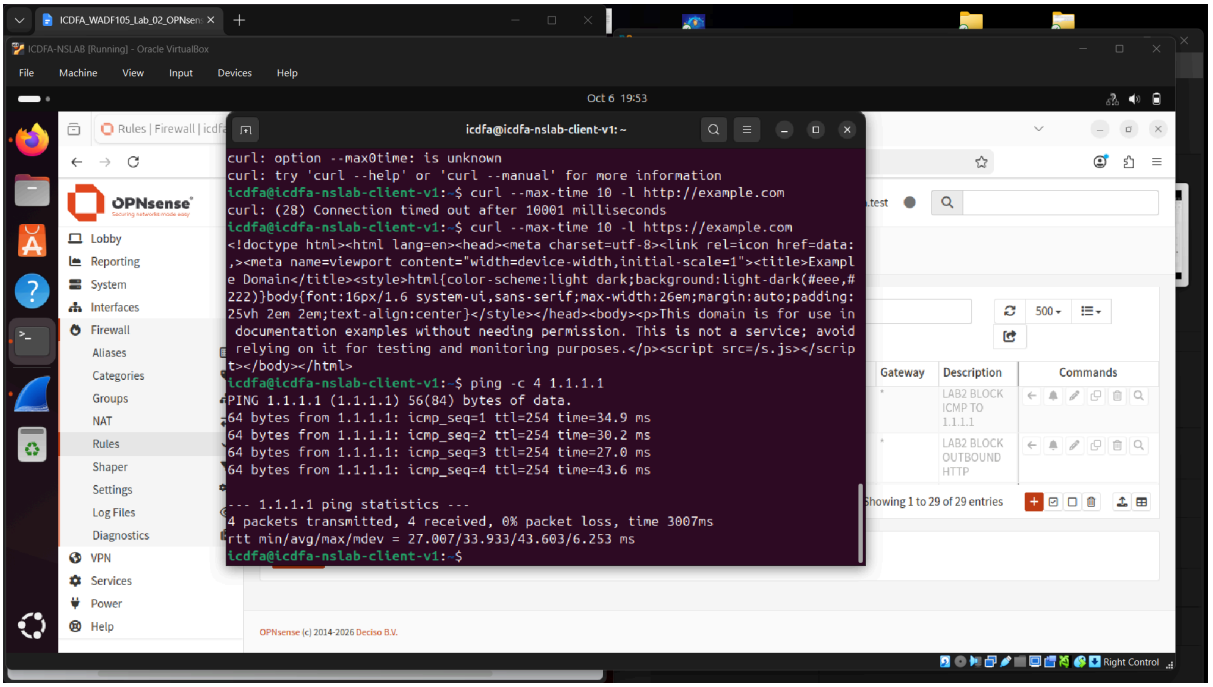

ICDFA_WADF105_Lab_02_OPNsen
https://docs.google.com/document/d/15xTRVttN...
ICDFA_WADF105_Lab_02_...
Plik Edytuj Widok Wstaw Formatuj ...
ICDFA-NSLAB [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Oct 6 19:48
*enp0s3
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.dstport ==80
No. Time Source Destination Protocol Length Info
556 89.291123674 10.10.10.195 185.125.190.99 TCP 74 44238 → 80
565 90.355596985 10.10.10.195 185.125.190.99 TCP 74 [TCP Retra
568 91.380165807 10.10.10.195 185.125.190.99 TCP 74 [TCP Retra
575 92.403040877 10.10.10.195 185.125.190.99 TCP 74 [TCP Retra
584 93.427084266 10.10.10.195 185.125.190.99 TCP 74 [TCP Retra
593 94.456045370 10.10.10.195 185.125.190.99 TCP 74 [TCP Retra
600 96.498571482 10.10.10.195 185.125.190.99 TCP 74 [TCP Retra
625 100.530599805 10.10.10.195 185.125.190.99 TCP 74 [TCP Retra
677 108.851388036 10.10.10.195 185.125.190.99 TCP 74 [TCP Retra
687 110.378851533 10.10.10.195 172.66.147.243 TCP 74 34726 → 80
698 111.410938926 10.10.10.195 172.66.147.243 TCP 74 [TCP Retra
705 112.462962293 10.10.10.195 172.66.147.243 TCP 74 [TCP Retra
708 113.522465268 10.10.10.195 172.66.147.243 TCP 74 [TCP Retra
Frame 556: 74 bytes on wire (592 bits), 74 by
Ethernet II, Src: PCSSystemtec_cd:8e:fb (08:6
Internet Protocol Version 4, Src: 10.10.10.19
Transmission Control Protocol, Src Port: 4423
0000 08 00 27 06 4d 55 08 00 27 cd 8e fb 08
0010 00 3c 09 f1 40 00 40 06 a4 1d 0a 0a 0a
0020 be 63 ac ce 00 50 f7 4c 78 67 00 00 00
0030 fa f0 8c dc 00 00 02 04 05 b4 04 02 08
0040 06 b2 00 00 00 00 01 03 03 0a
wireshark_enp0s39ZO6W3.pcapng
Packets: 1249 · Displayed: 84 (6.7%) · Dropped: 0 (0.0%)
Profile: Default
Right Control
WADF105 | Pr
20. Use the filters below to isolate the traffic.
Display filter
Expected observation
icmp && ip.addr == 1.1.1.1
Echo requests leave Ubuntu, but no echo r
tcp.dstport == 80
TCP SYN packets may be retransmitted be
firewall silently blocks them.
tcp.port == 443
A permitted HTTPS connection shows a co
handshake and encrypted TLS traffic.
The exact retransmission count may vary. Focus on the difference between blocked traffic wi
successful response and permitted traffic that completes a connection.
Part G Inspect states and automatic NAT
21. Open Firewall, Diagnostics, States and filter by the icdfa-nslab-client-v1 IP address.
22. Generate a fresh HTTPS request from icdfa-nslab-client-v1 and refresh the state table.
23. Identify the LAN-side connection and its translated WAN-side state.
24. Open Firewall, NAT, Outbound and confirm the current mode. In a standard installation, a
outbound NAT translates the 10.10.10.0/24 private source address for internet access.
25. Do not change outbound NAT mode during this laboratory.
Part H Restore the laboratory
26. Return to Firewall, Rules, LAN.
27. Disable the two rules named LAB2 BLOCK ICMP TO 1.1.1.1 and LAB2 BLOCK OUTBOU
not delete them until your screenshots are complete.
28. Apply the changes.

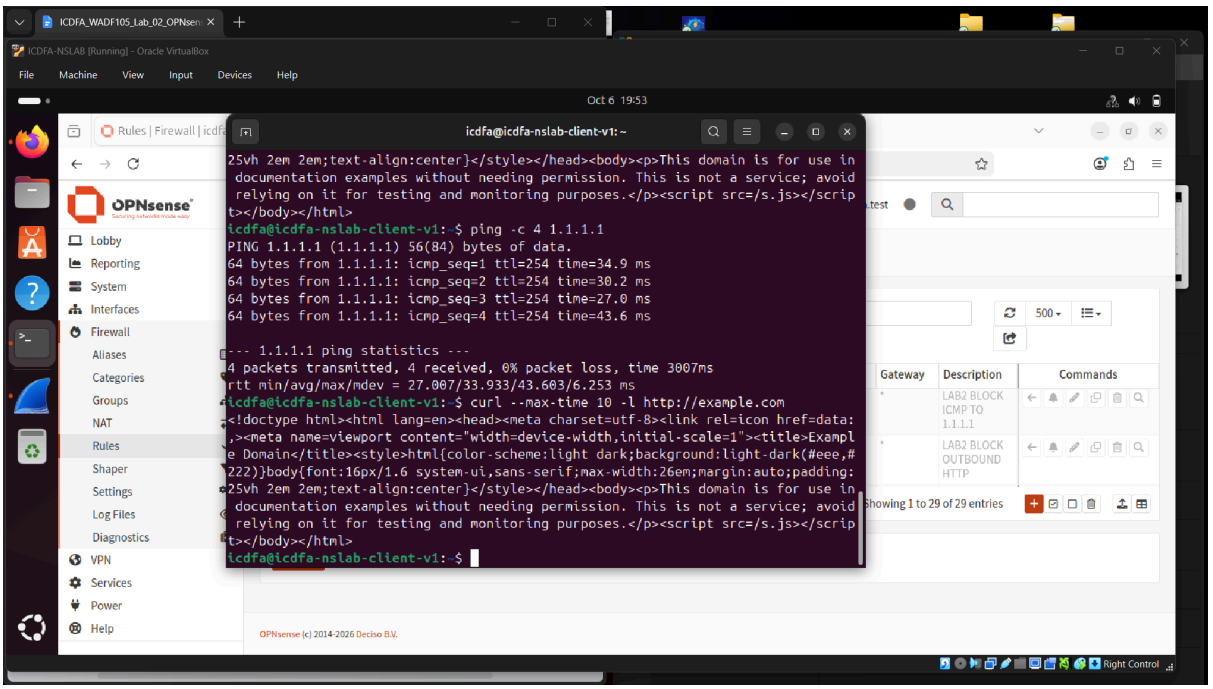

ICDFA_WADF105_Lab_02_OPNsen
https://docs.google.com/document/d/15xTRVttN...
ICDFA_WADF105_Lab_02_...
Plik Edytuj Widok Wstaw Formatuj ...
ICDFA-NSLAB [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Oct 6 19:48
*enp0s3
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.port ==443
No. Time Source Destination Protocol Length Info
548 88.125881735 10.10.10.1 10.10.10.195 TLSv1.2 97 Applicatio
549 88.125973595 10.10.10.195 10.10.10.1 TCP 66 34096 → 44
550 89.068897695 10.10.10.1 10.10.10.195 TLSv1.2 188 Applicatio
551 89.011089132 10.10.10.1 10.10.10.195 TCP 66 443 → 3409
552 89.146980941 10.10.10.1 10.10.10.195 TLSv1.2 932 Applicatio
553 89.147037226 10.10.10.195 10.10.10.1 TCP 66 34096 → 44
554 89.149886644 10.10.10.1 10.10.10.195 TLSv1.2 97 Applicatio
555 89.149908727 10.10.10.195 10.10.10.1 TCP 66 34096 → 44
557 90.022631489 10.10.10.195 10.10.10.1 TLSv1.2 188 Applicatio
558 90.025552259 10.10.10.1 10.10.10.195 TCP 66 443 → 3409
559 90.144728113 10.10.10.1 10.10.10.195 TCP 1514 443 → 3409
560 90.144842687 10.10.10.195 10.10.10.1 TCP 66 34096 → 44
561 90.146077506 10.10.10.1 10.10.10.195 TLSv1.2 360 Applicatio
Frame 555: 66 bytes on wire (528 bits), 66 by
Ethernet II, Src: PCSSystemtec_cd:8e:fb (08:6
Internet Protocol Version 4, Src: 10.10.10.19
Transmission Control Protocol, Src Port: 3409
0000 08 00 27 06 4d 55 08 00 27 cd 8e fb 08
0010 00 34 d0 7e 40 00 40 06 41 6e 0a 0a 0a
0020 0a 01 85 30 01 bb 37 d7 66 76 3f ea 85
0030 00 e8 28 fe 00 00 01 01 08 0a 23 02 40
0040 80 d5
wireshark_enp0s39ZO6W3.pcapng
Packets: 1249 · Displayed: 1153 (92.3%) · Dropped: 0 (0.0%)
Profile: Default
Right Control
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Display filter
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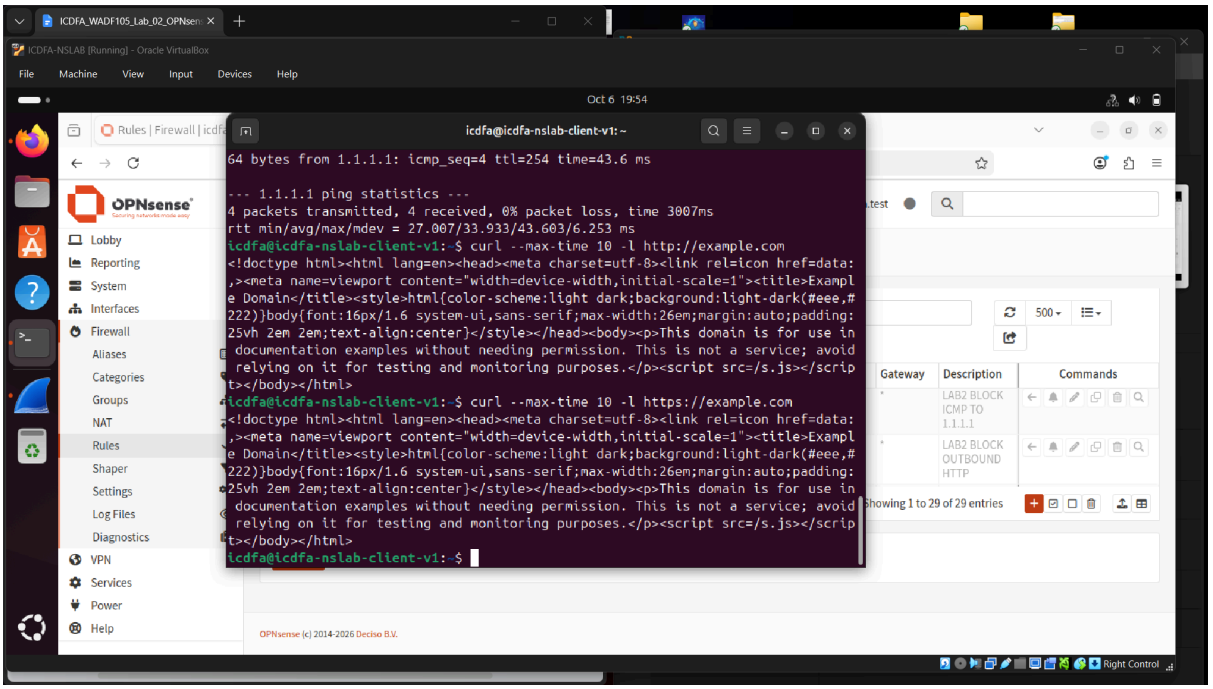
Rules disabled:



Running commands resulting in success:







Questions answers:

1. Why must the specific block rules be placed above the broad allow rule?

OPNsense processes firewall rules sequentially from top to bottom. If a specific block rule is placed below the broad "Default allow LAN to any" rule, the firewall will match the traffic to the allow rule first and permit it before it ever reaches the block rule. Putting the block rules at the top ensures they are evaluated first.

2. Which five packet attributes are most useful when explaining a firewall decision?

The five most useful packet attributes are:

- ***Protocol*** *(e.g., ICMP, TCP, UDP)*
- ***Source IP Address***
- ***Destination IP Address***
- ***Destination Port*** *(e.g., Port 80 for HTTP, Port 443 for HTTPS)*
- ***Interface / Direction*** *(e.g., LAN, Inbound)*

3. Why did blocking ICMP not block HTTPS?

Blocking ICMP did not block HTTPS because they use entirely different network protocols. The firewall rule specifically targeted ICMP packets (which are used by the `ping` *command), while HTTPS traffic uses the TCP protocol on port 443. Since the rule didn't match TCP traffic, HTTPS was allowed through normally.*

4. What difference did you observe between the blocked TCP port 80 traffic and permitted TCP port 443 traffic?

For the blocked ***TCP port 80 (HTTP)*** *traffic, the client sent* `TCP SYN` *packets, but the firewall silently dropped them. This caused repeated retransmissions from the client and resulted in a request timeout.*

For the permitted ***TCP port 443 (HTTPS)*** *traffic, the connection successfully completed a full 3-way TCP handshake (*`SYN, SYN-ACK, ACK`*) followed by encrypted TLS application data exchange.*

5. What role does outbound NAT play when icdfa-nslab-client-v1 uses a private IPv4 address?

Outbound NAT translates the internal client's private IP address (`10.10.10.x`*) into the firewall's public/WAN IP address as packets leave the network. Because private IP addresses cannot be routed over the public internet, NAT allows external web servers to route response traffic back to the firewall's WAN interface, which then forwards it back to the client machine.*

6. Why is restoring the original state an important part of a controlled security laboratory?

Restoring the original baseline configuration ensures that temporary test rules do not interfere with network connectivity or distort the outcomes of future lab exercises. It keeps the testing environment clean, predictable, and ready for subsequent configurations.